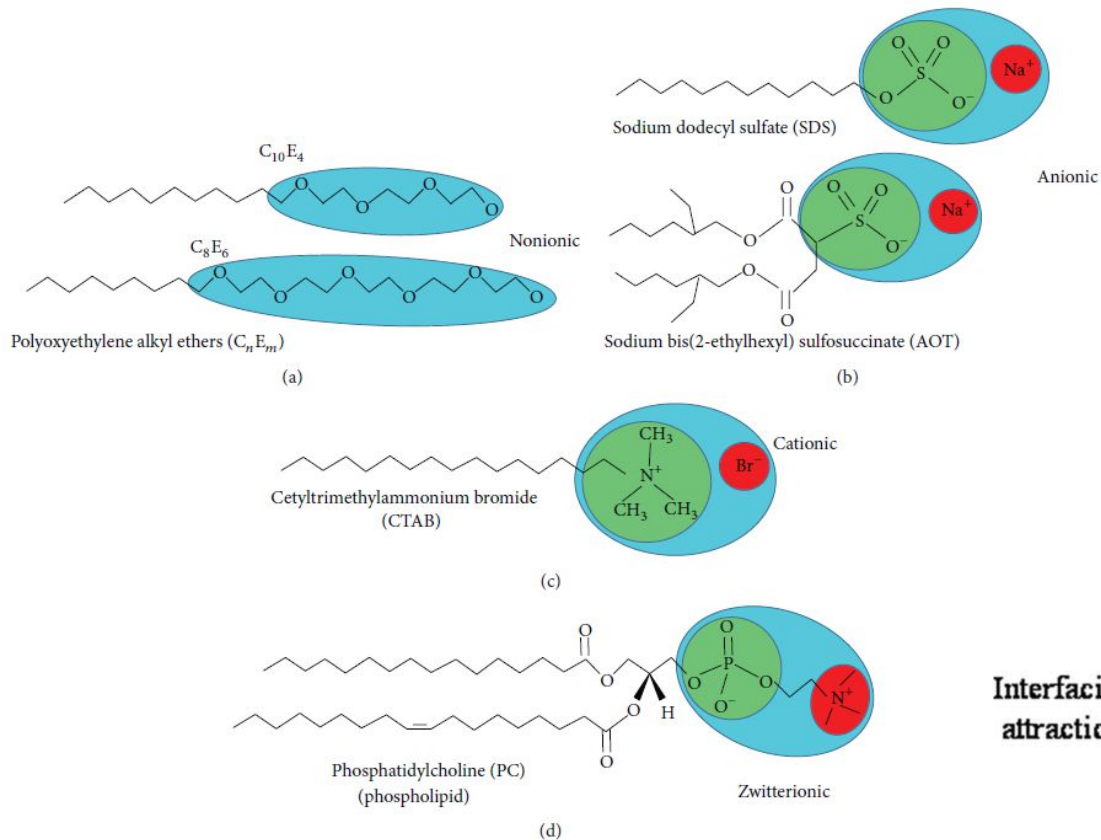


(Purely) Synthetic Block Copolymers
- Materials from Solution Self-Assembly -

(Purely) Synthetic Block Copolymers
- Materials from Solution Self-Assembly -
Generalities

Theoretical Aspects of Solution Self-Assembly

Packing Parameter (p , also called shape factor)



$$p = \frac{v}{a_0 l_c}$$

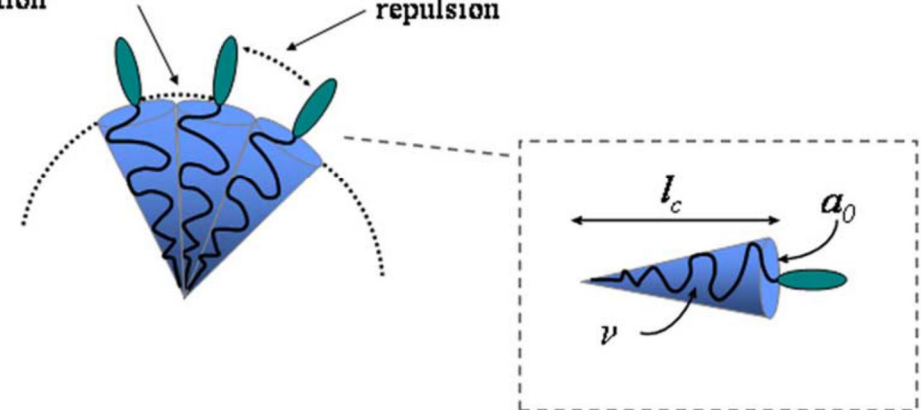
v : volume of the hydrophobic segment

a_0 : contact area of the head group

l_c : length of the hydrophobic segment

Interfacial (hydrophobic)
attraction

Head-group (hydrophilic)
repulsion

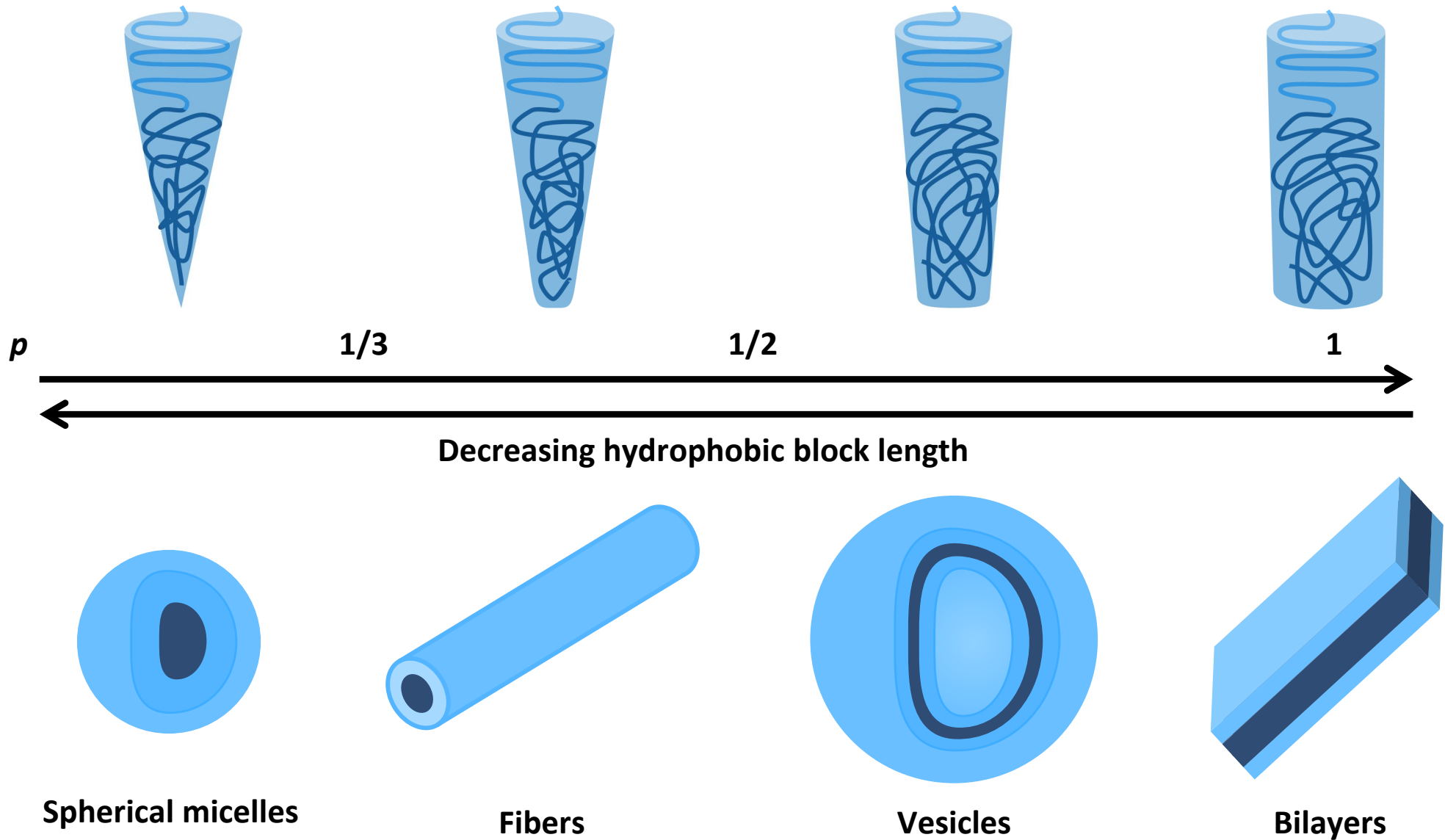


Advances in Condensed Matter Physics Volume 2015, Article ID 151683

J. Israelachvili, "Intermolecular & Surface Forces", 2nd ed., Academic Press, London **1991**.

Theoretical Aspects of Solution Self-Assembly

Packing Parameter



Science 2002, 297, 967

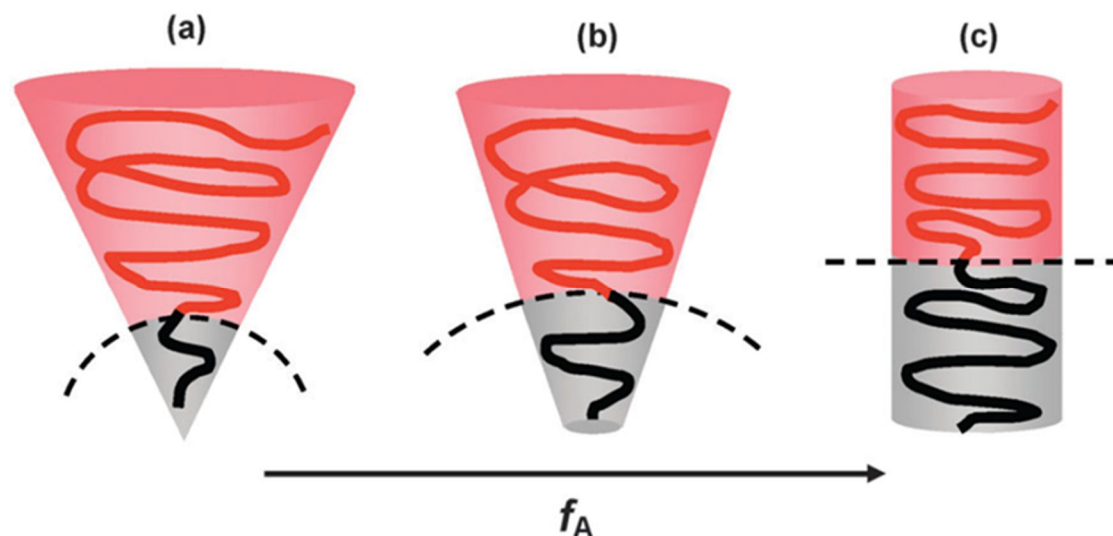


Fig. 3 Schematic illustration of the possible polymer chain arrangements in different morphologies of AB diblocks changing from sphere (a) to cylinder (b) and to lamella (c), as the volume fraction (f_A) of the A block (black) increases to ~ 0.5 .

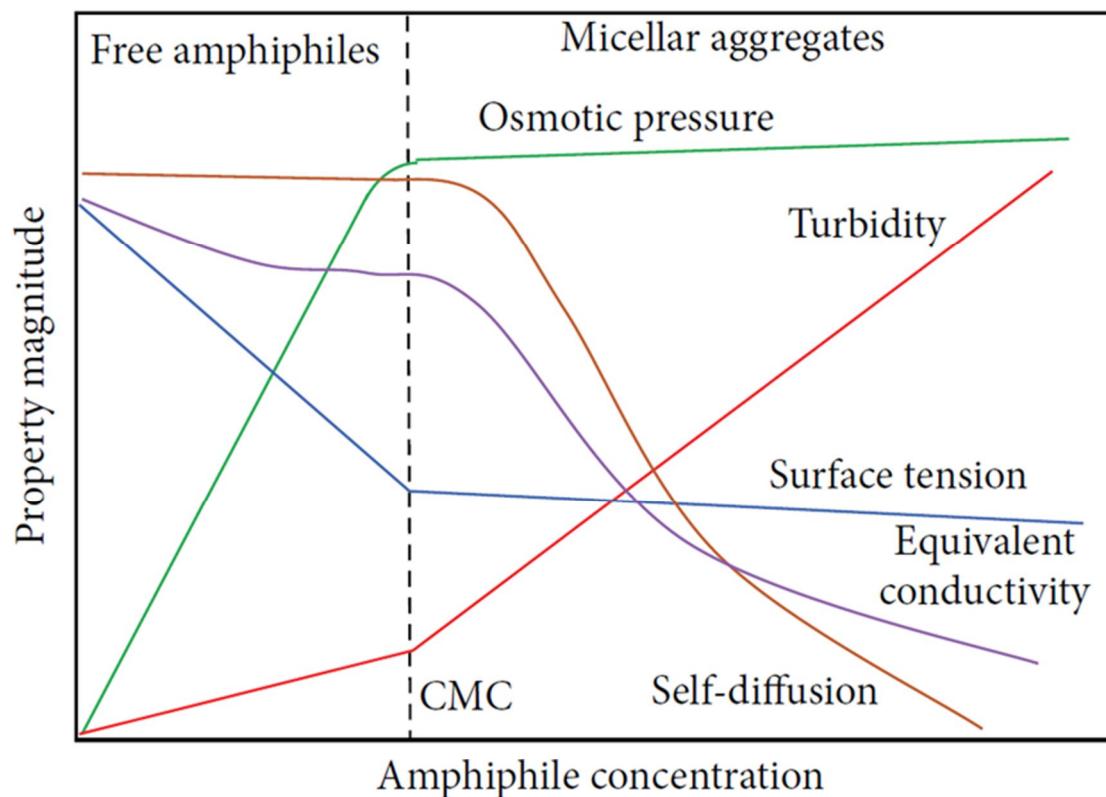


FIGURE 4: Behavior of some physical properties and their discontinuity associated with the formation of micelles at the critical micelle concentration (CMC) for amphiphiles in a selective solvent.

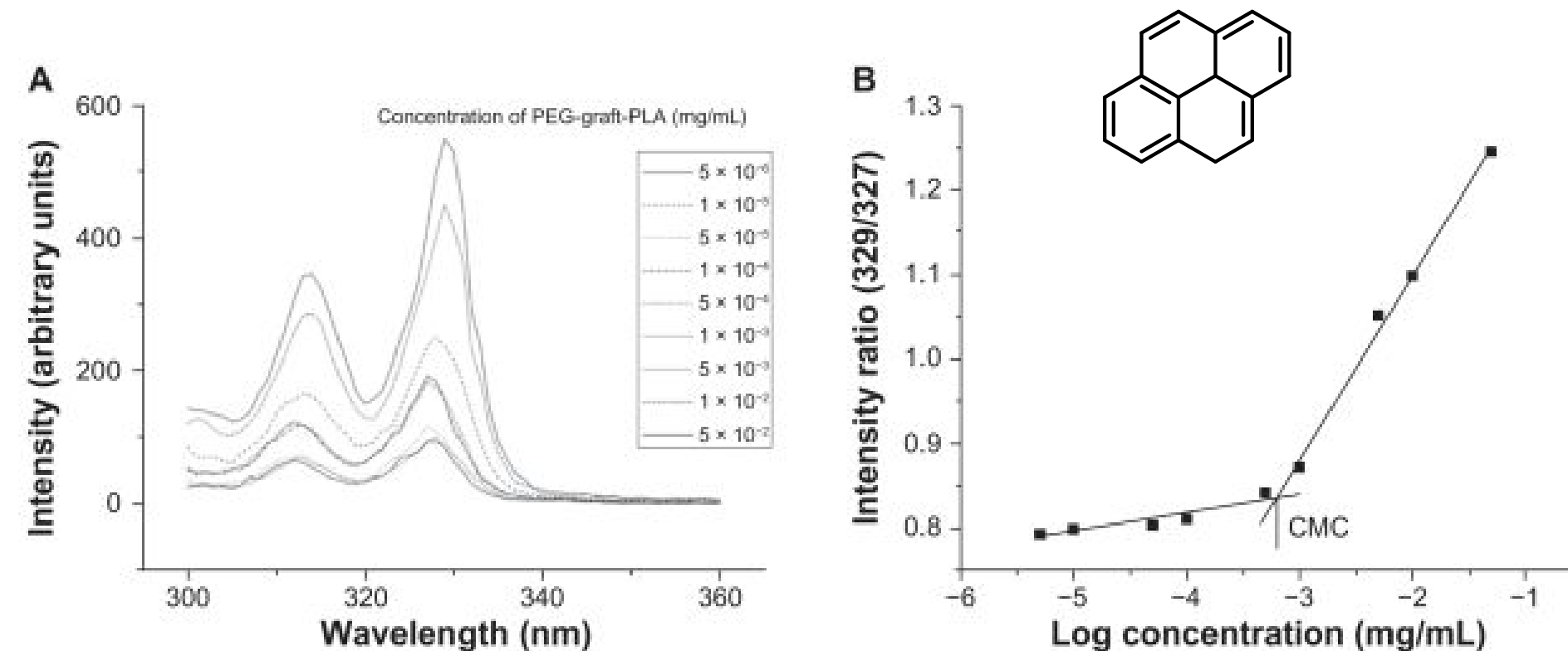
Theoretical Aspects of Solution Self-Assembly

Critical Micellar Concentration (CMC)

Experimental Determination

The CMC of block copolymers is very low, as compared to that of small amphiphiles

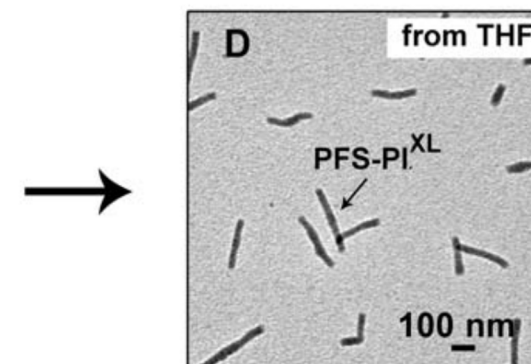
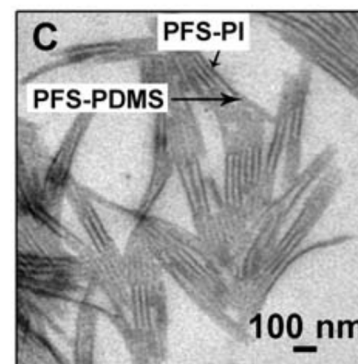
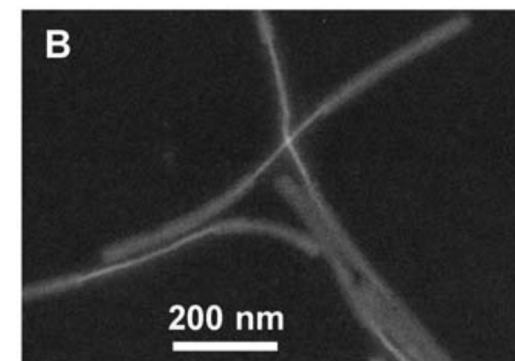
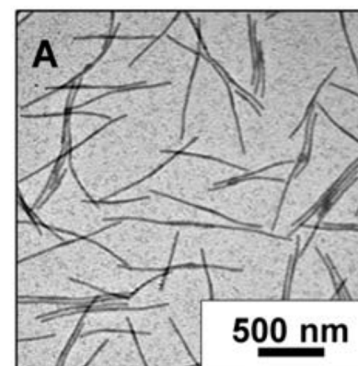
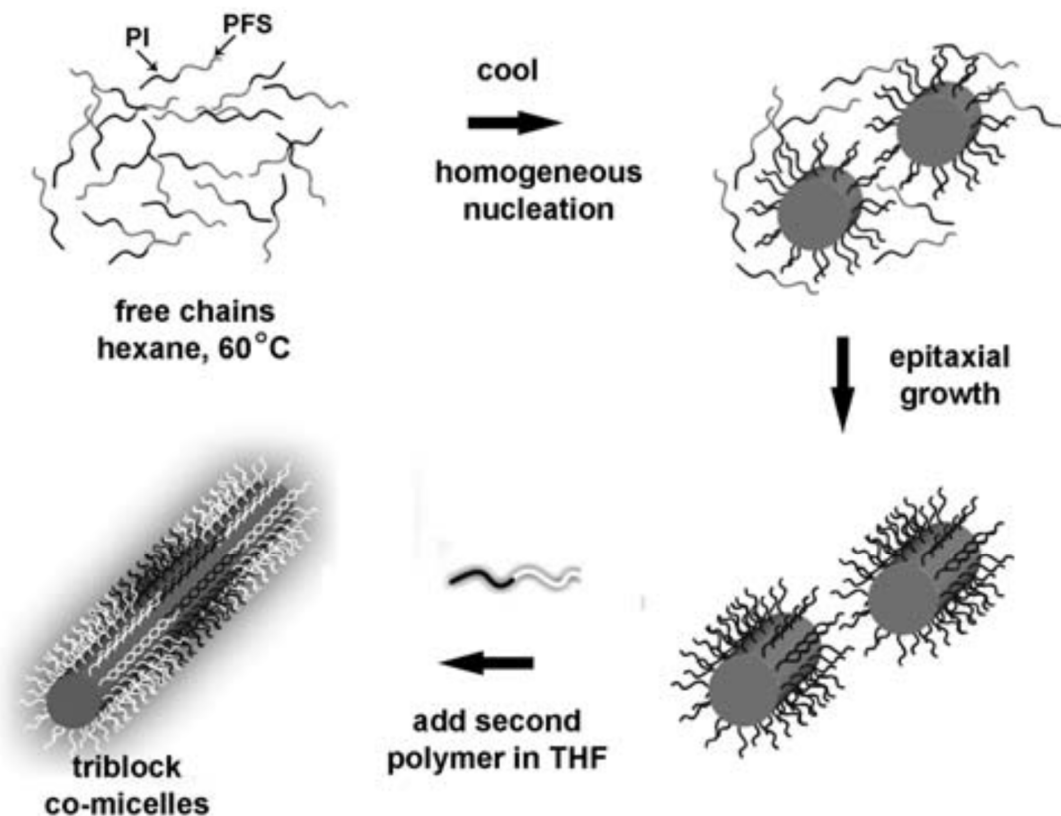
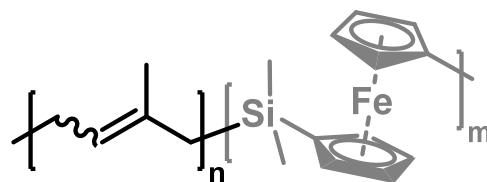
→ Requires very sensitive technique → fluorescence



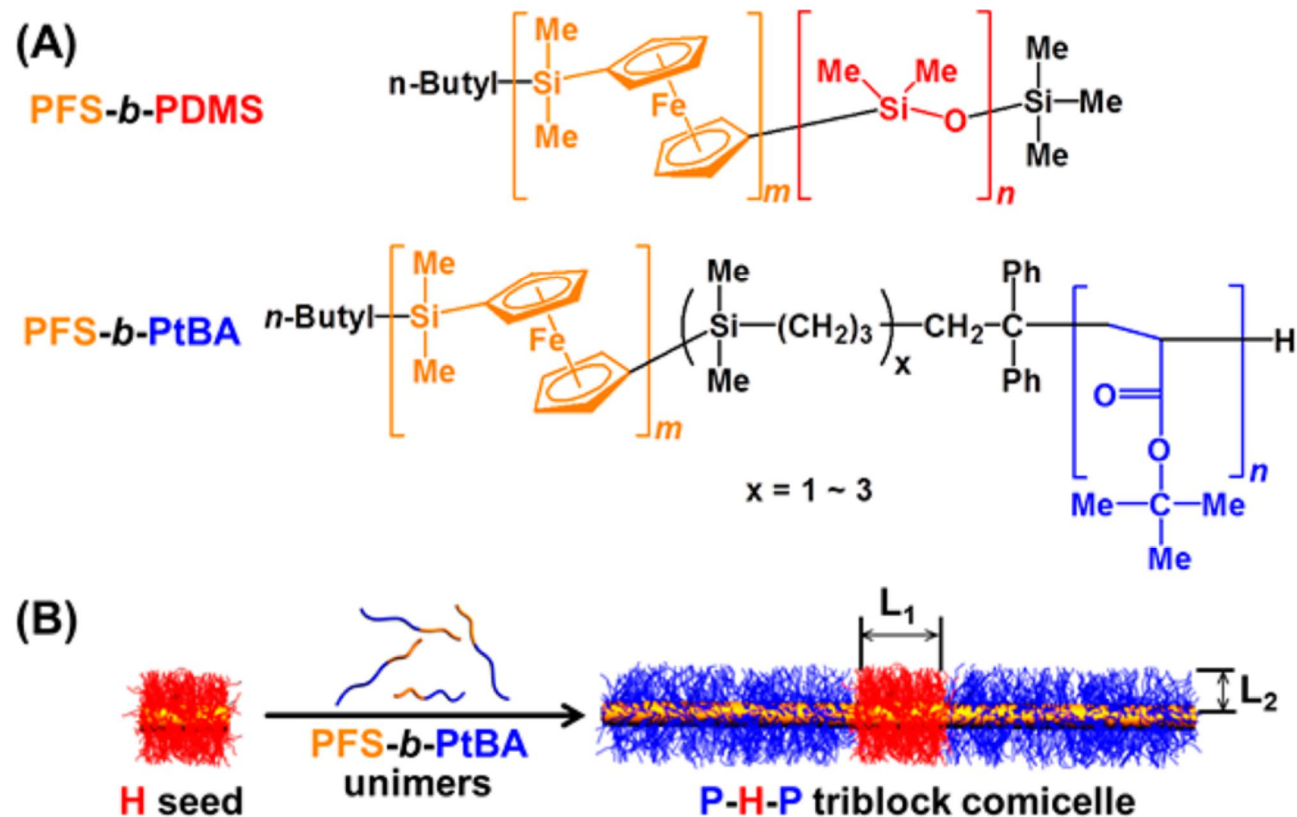
J. Am. Chem. Soc. **1977**, 99, 2039
Int. J. Nanomedicine **2011**, 6, 1443

Multicompartmented Nanoobjects

Co-Assembly of Micelles

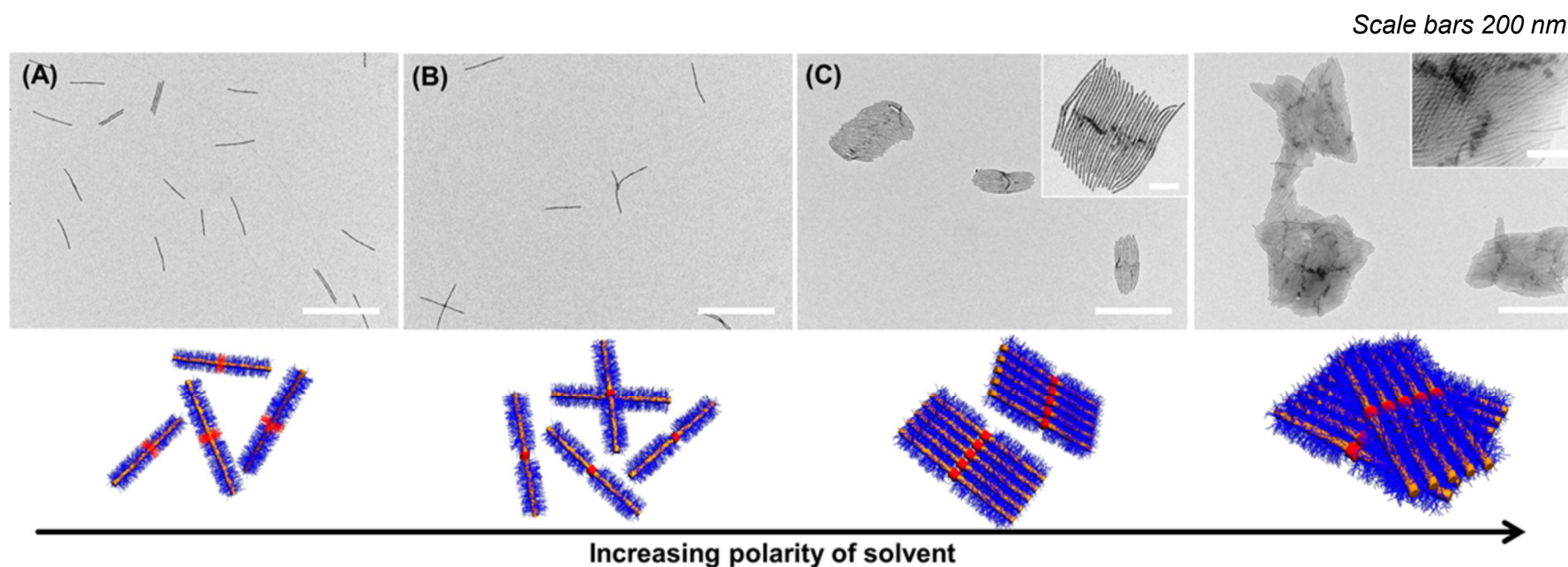


Science 2007, 317, 644



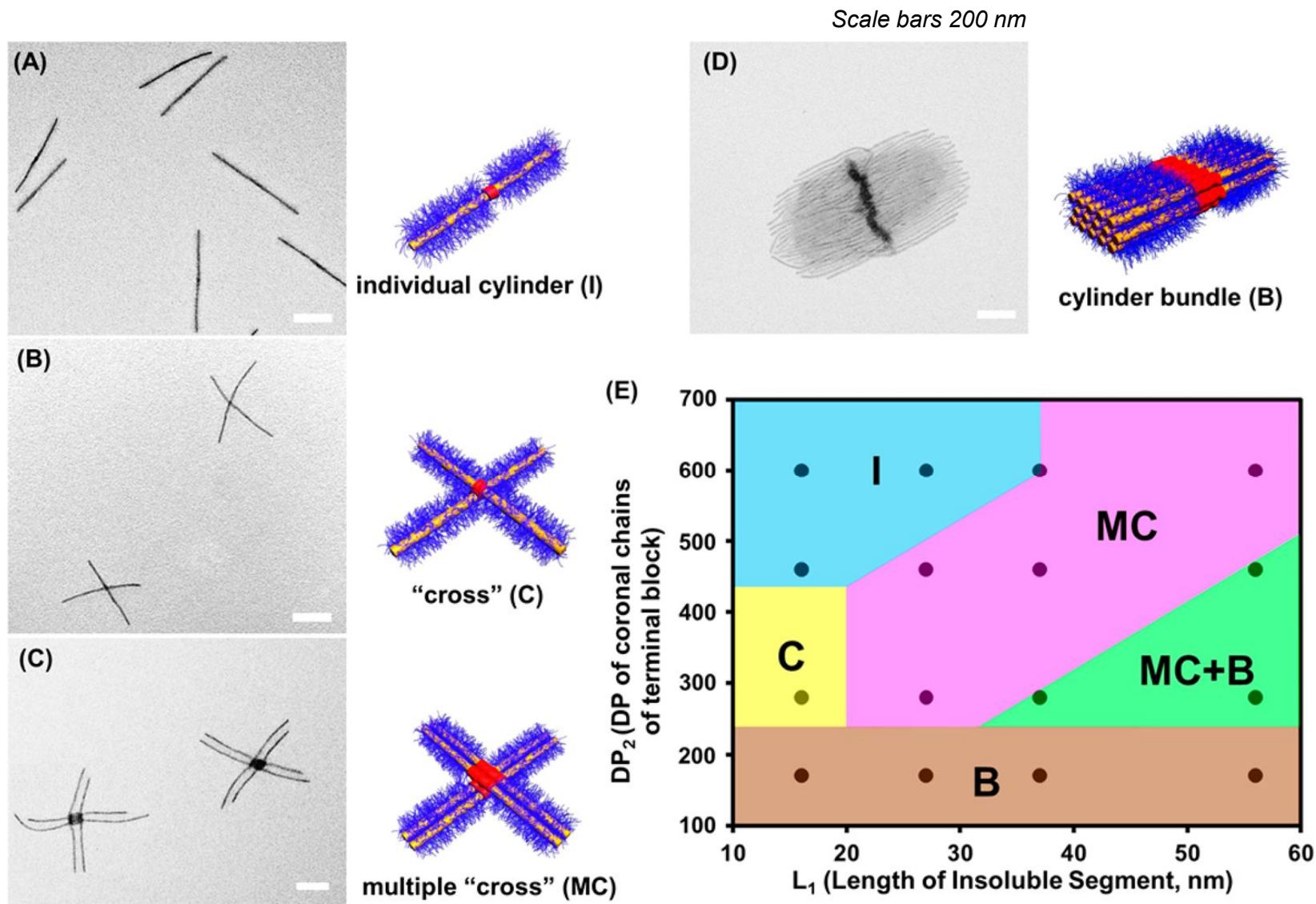
Multicompartmented Nanoobjects

Co-Assembly of Micelles



Multicompartmented Nanoobjects

Co-Assembly of Micelles

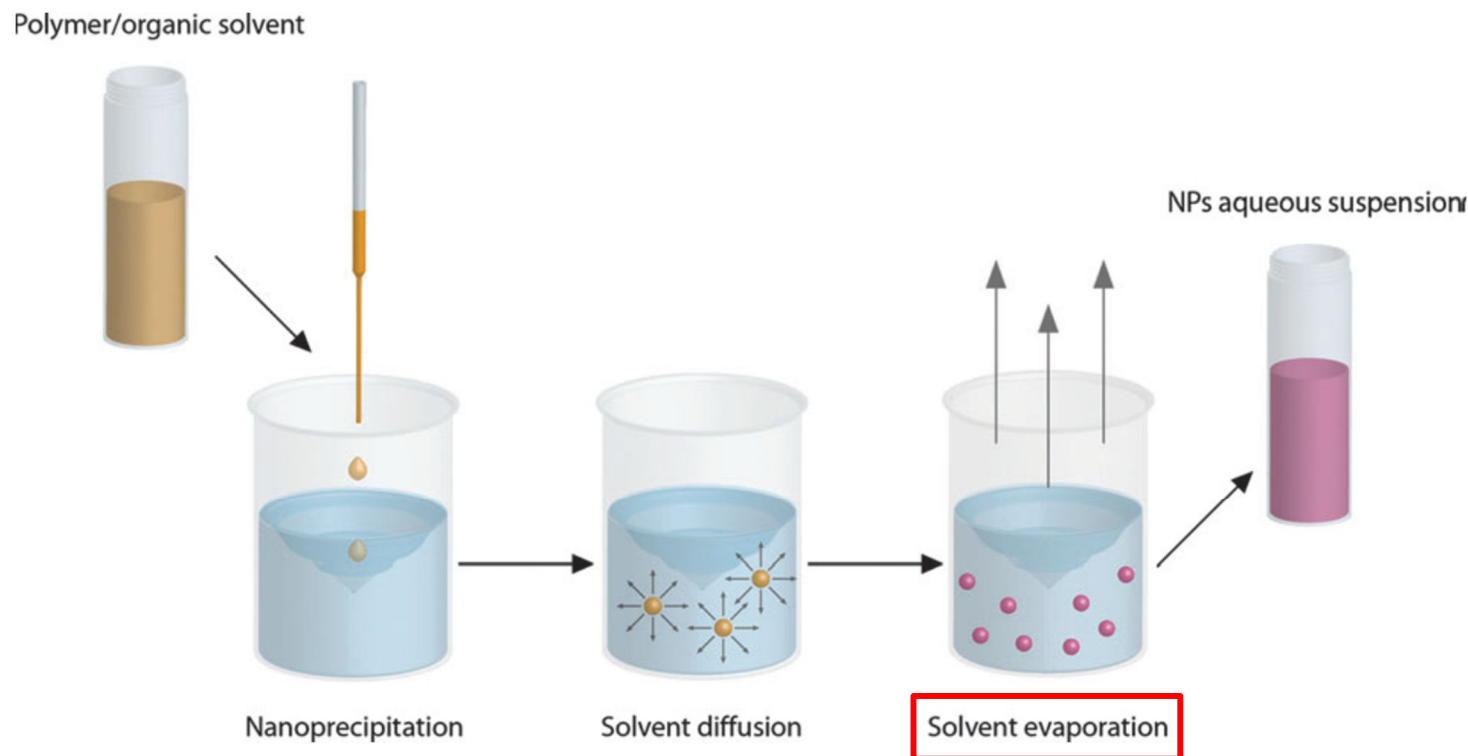


J. Am. Chem. Soc. **2016**, 138, 4087

(Purely) Synthetic Block Copolymers
- Materials from Solution Self-Assembly -
Preparation

Production of Nanoobjects

Co-Solvent Method (Nanoprecipitation)



Dialysis can be used alternatively
(e.g., for high boiling point solvent)

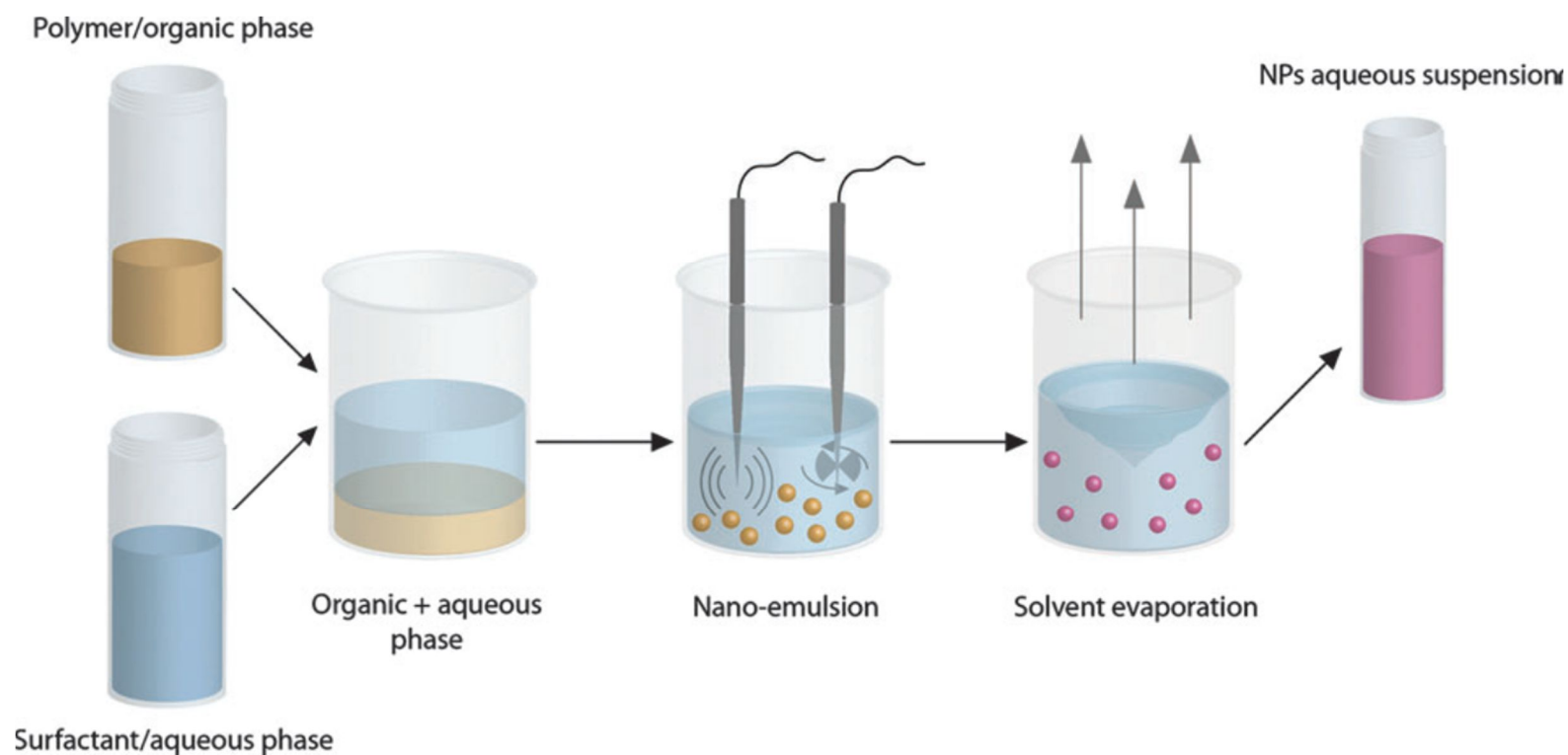
2 possible variations

- + The selective solvent is added dropwise to the solution of BCP in the non-selective solvent.
- + The non-selective solvent is not miscible and can evaporate fast (dichloromethane).

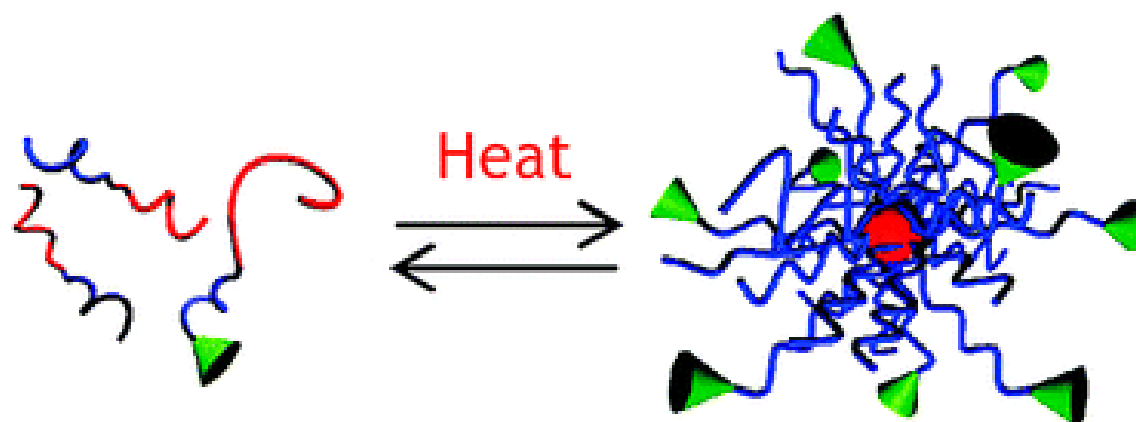
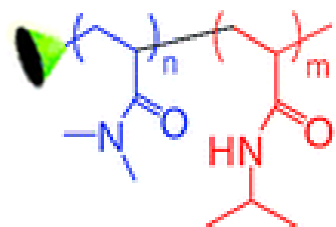
Chem. Soc. Rev. 2013, 42, 1147

Production of Nanoobjects

(Mini)emulsion



Chem. Soc. Rev. 2013, 42, 1147

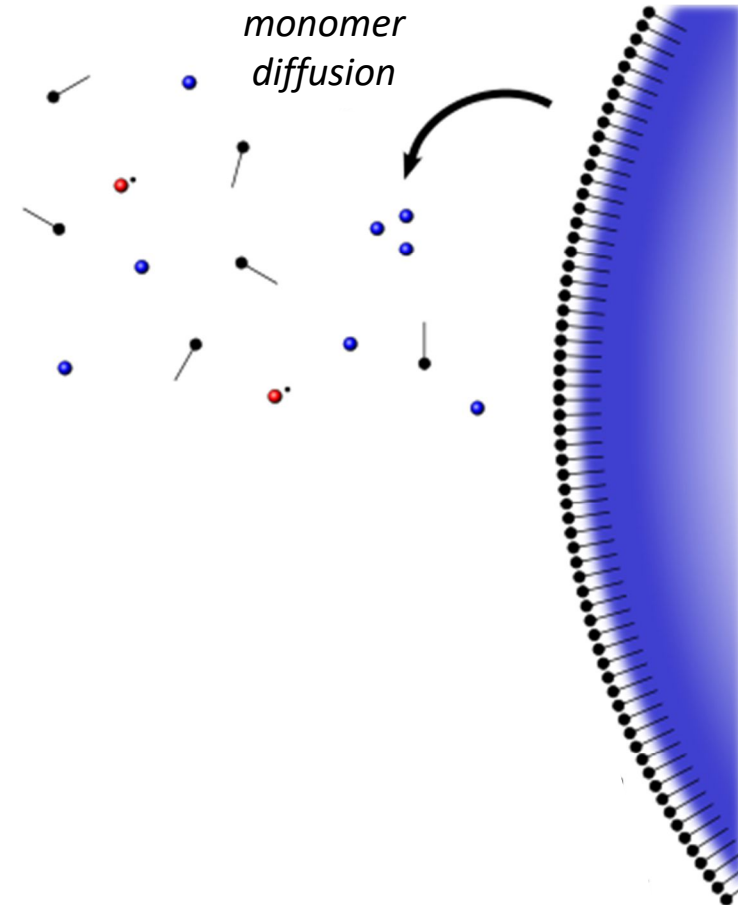
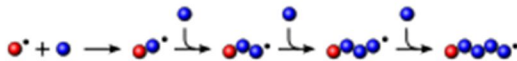


Production of Nanoobjects

Polymerization-Induced Self-Assembly

Emulsion polymerization mechanism

Oligoradical formation in aqueous phase

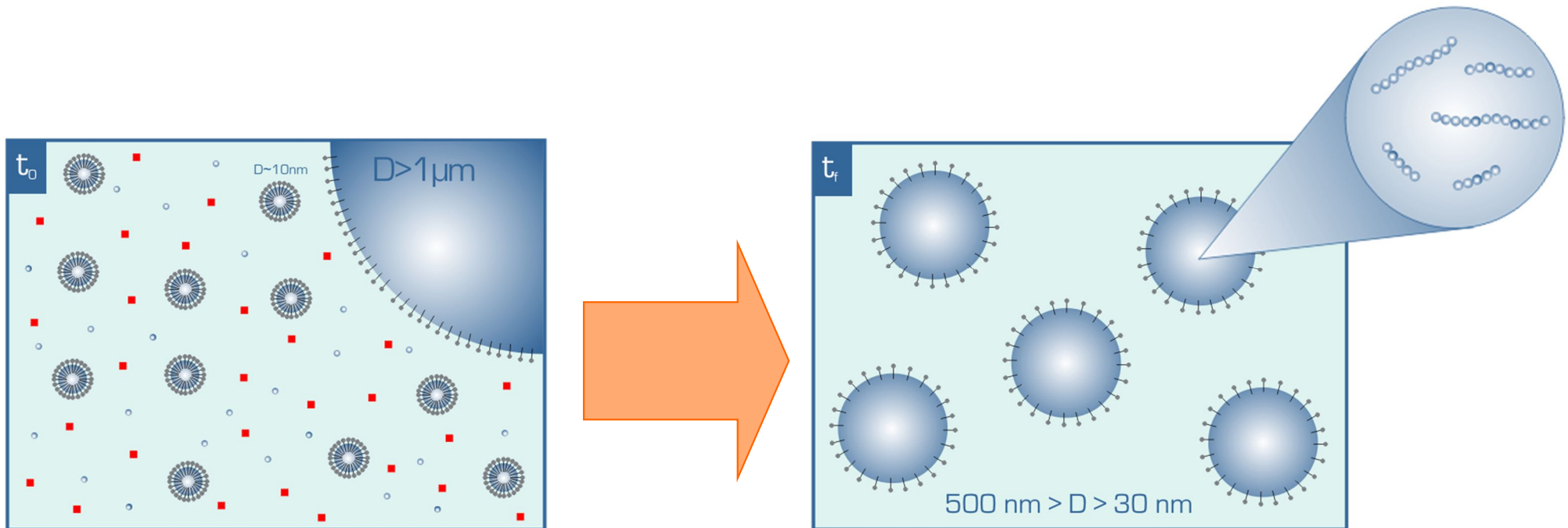


- surfactant
- monomer
- Initiating radical

Production of Nanoobjects

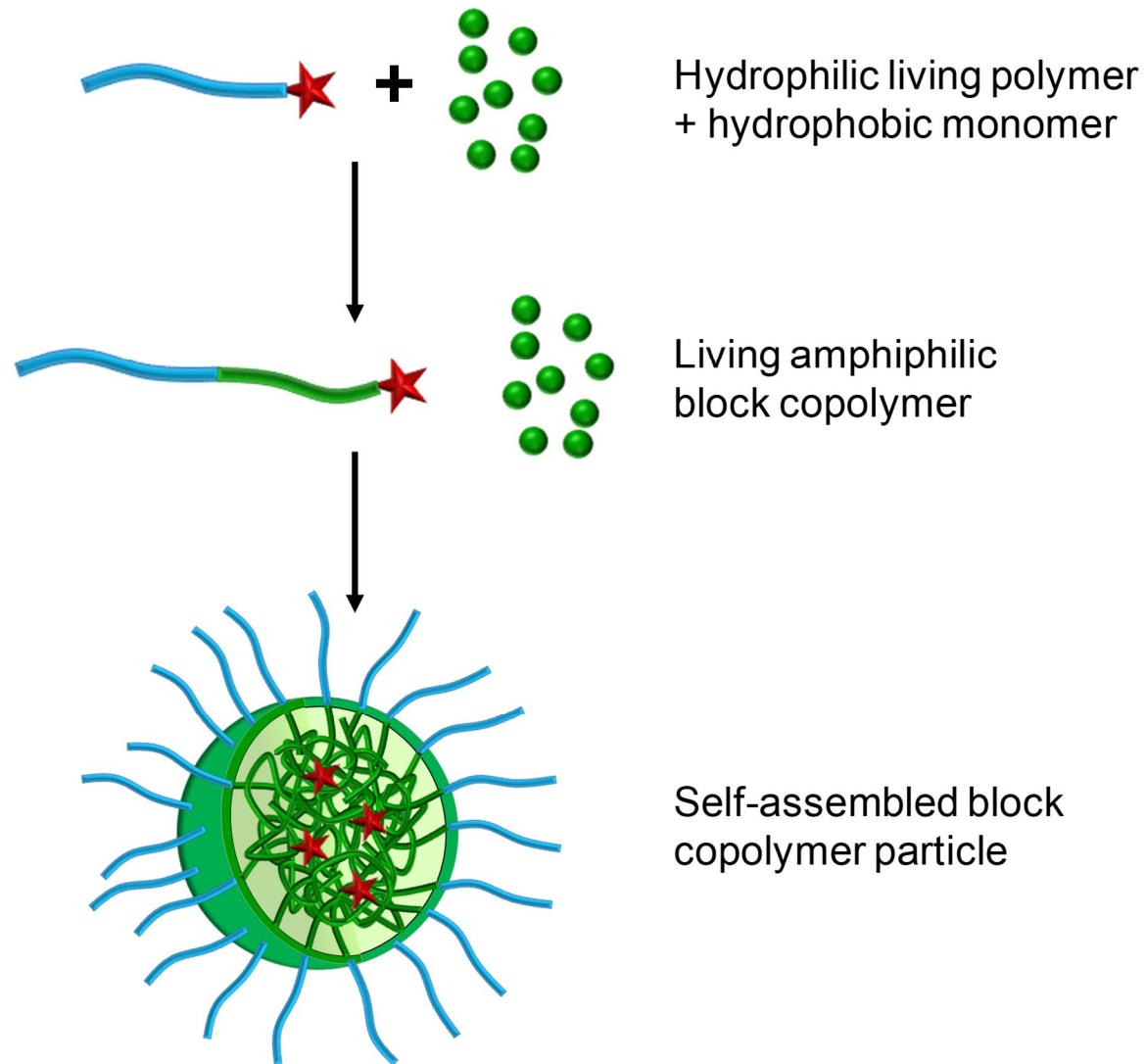
Polymerization-Induced Self-Assembly

Emulsion polymerization mechanism



Production of Nanoobjects

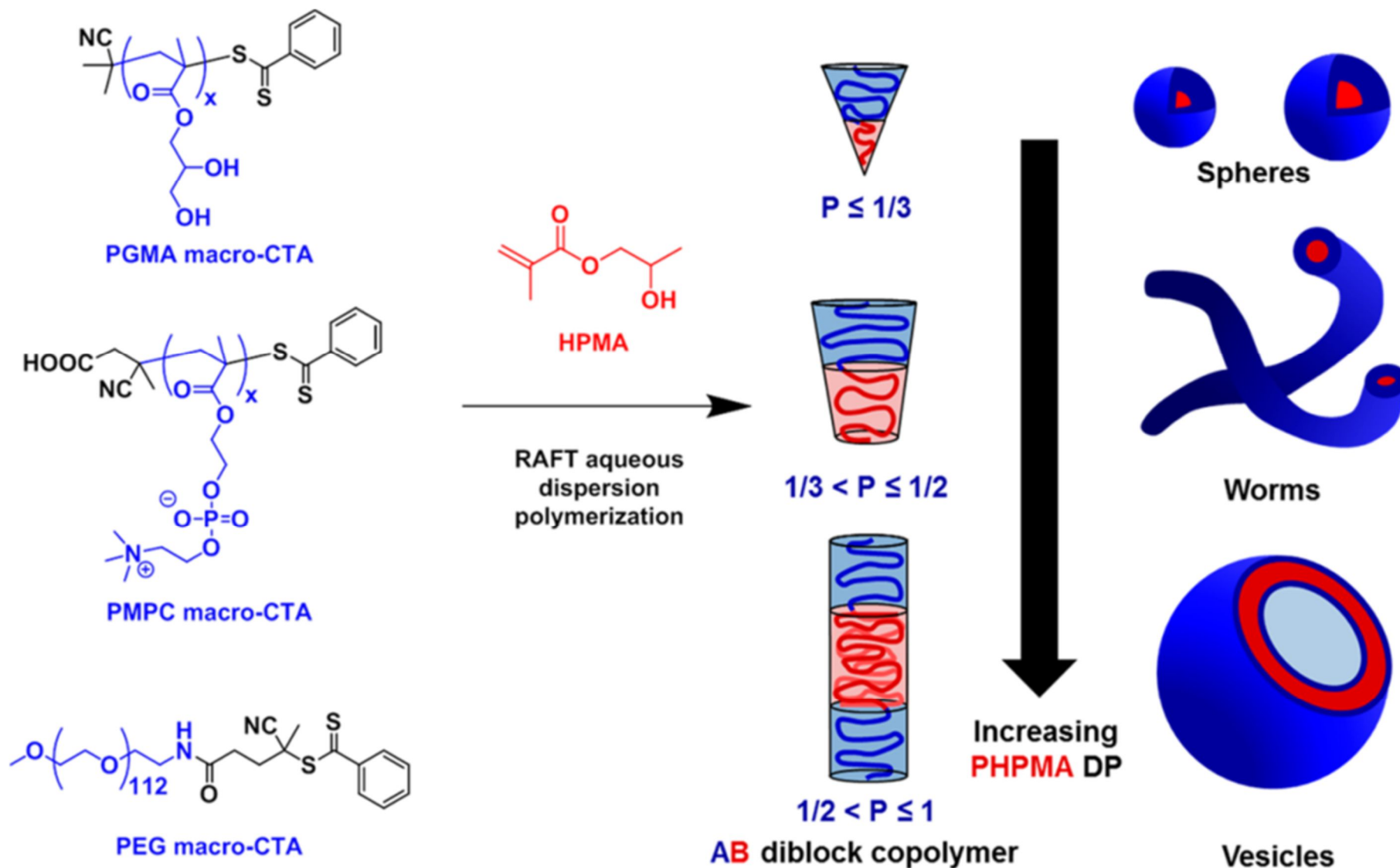
Polymerization-Induced Self-Assembly



Macromolecules **2012**, 45, 6753

Production of Nanoobjects

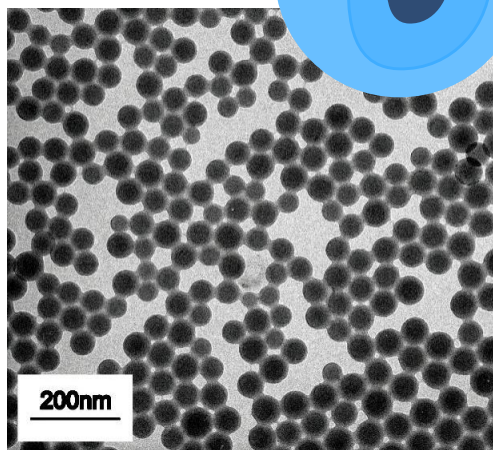
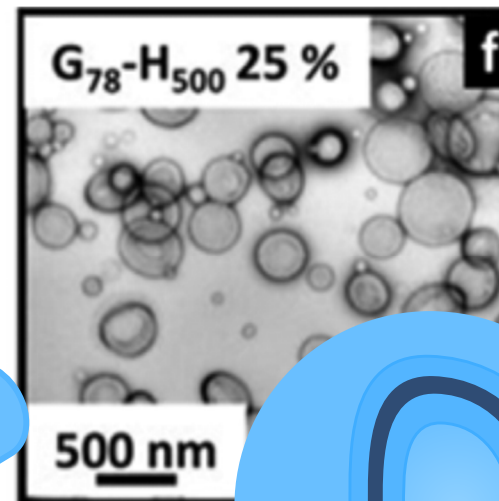
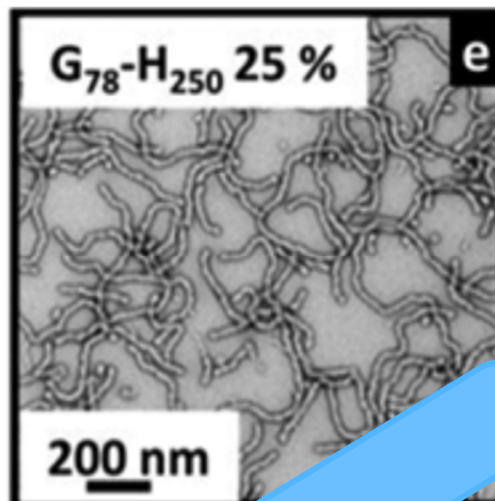
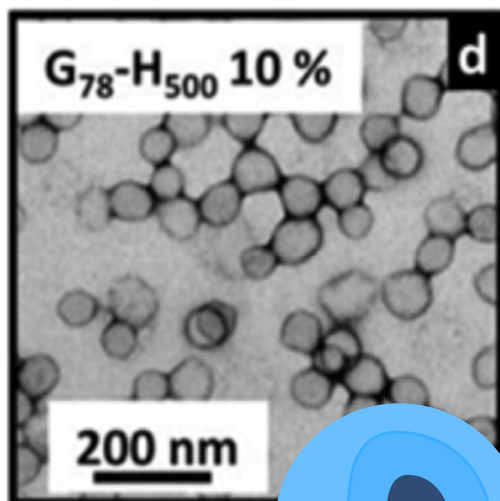
Polymerization-Induced Self-Assembly



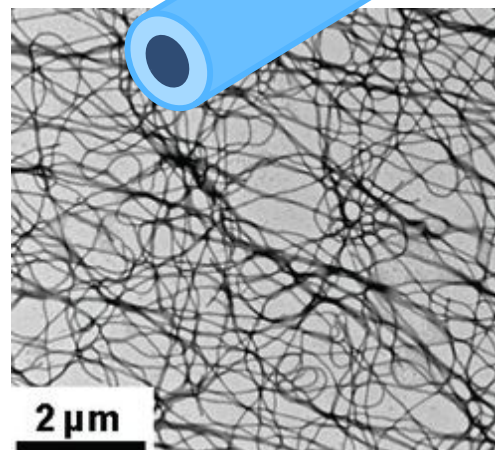
Production of Nanoobjects

Polymerization-Induced Self-Assembly

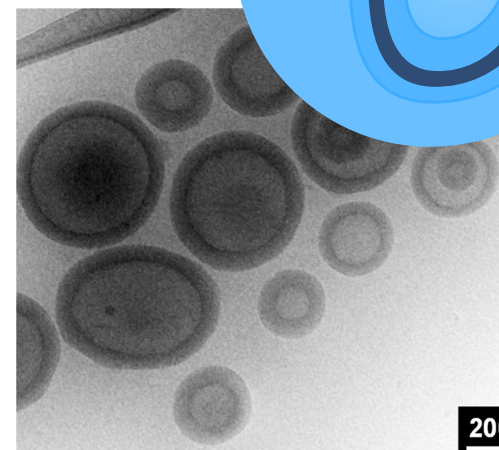
J. Am. Chem. Soc. **2014**, 136, 10174



Delaittre *et al.*
Chem. Commun. **2005**, 614



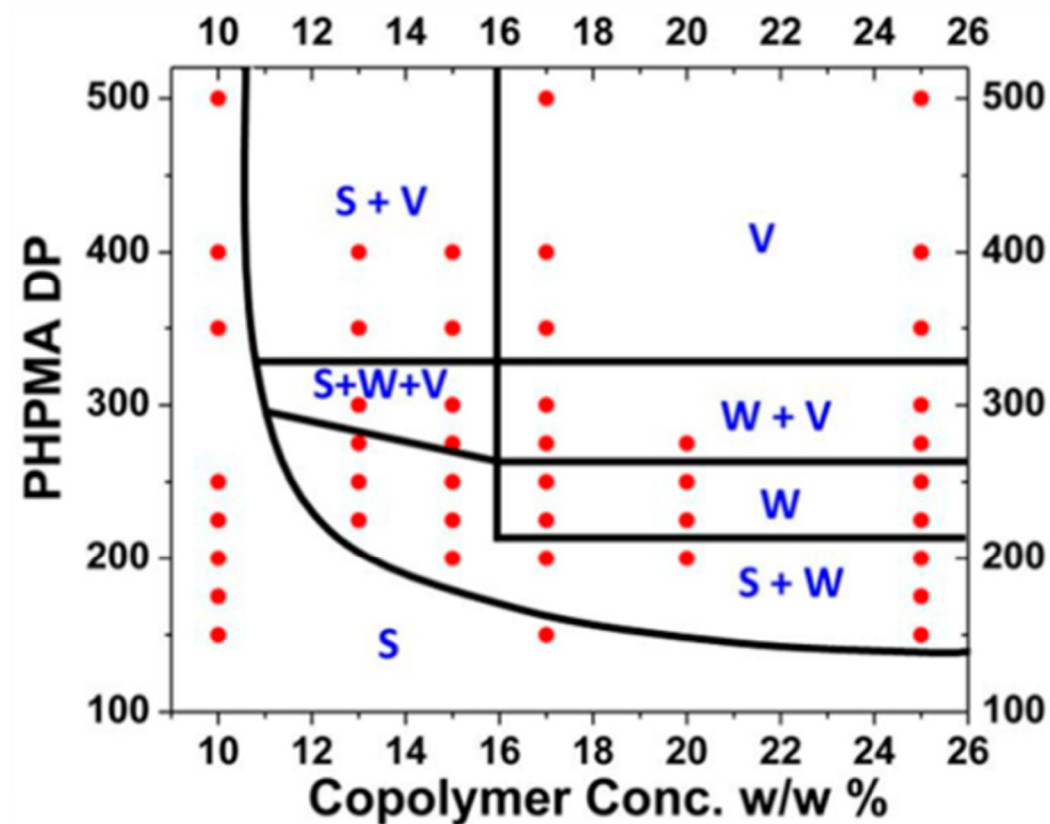
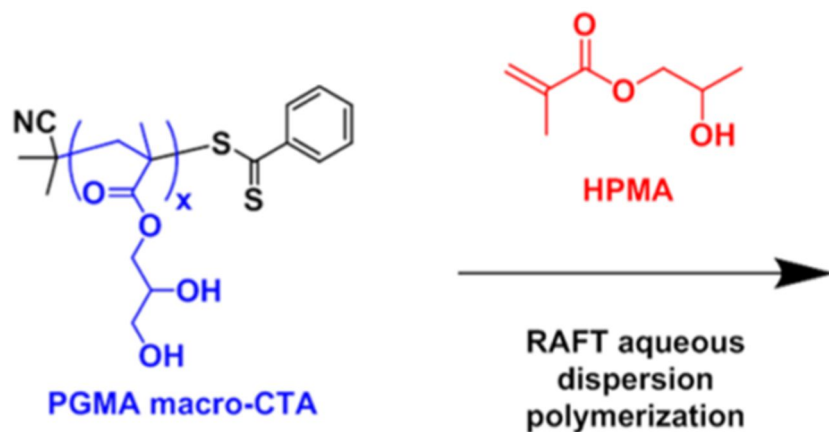
ACS Macro Lett. **2012**, 1, 47



Delaittre *et al.*
Chem. Commun. **2009**, 2887

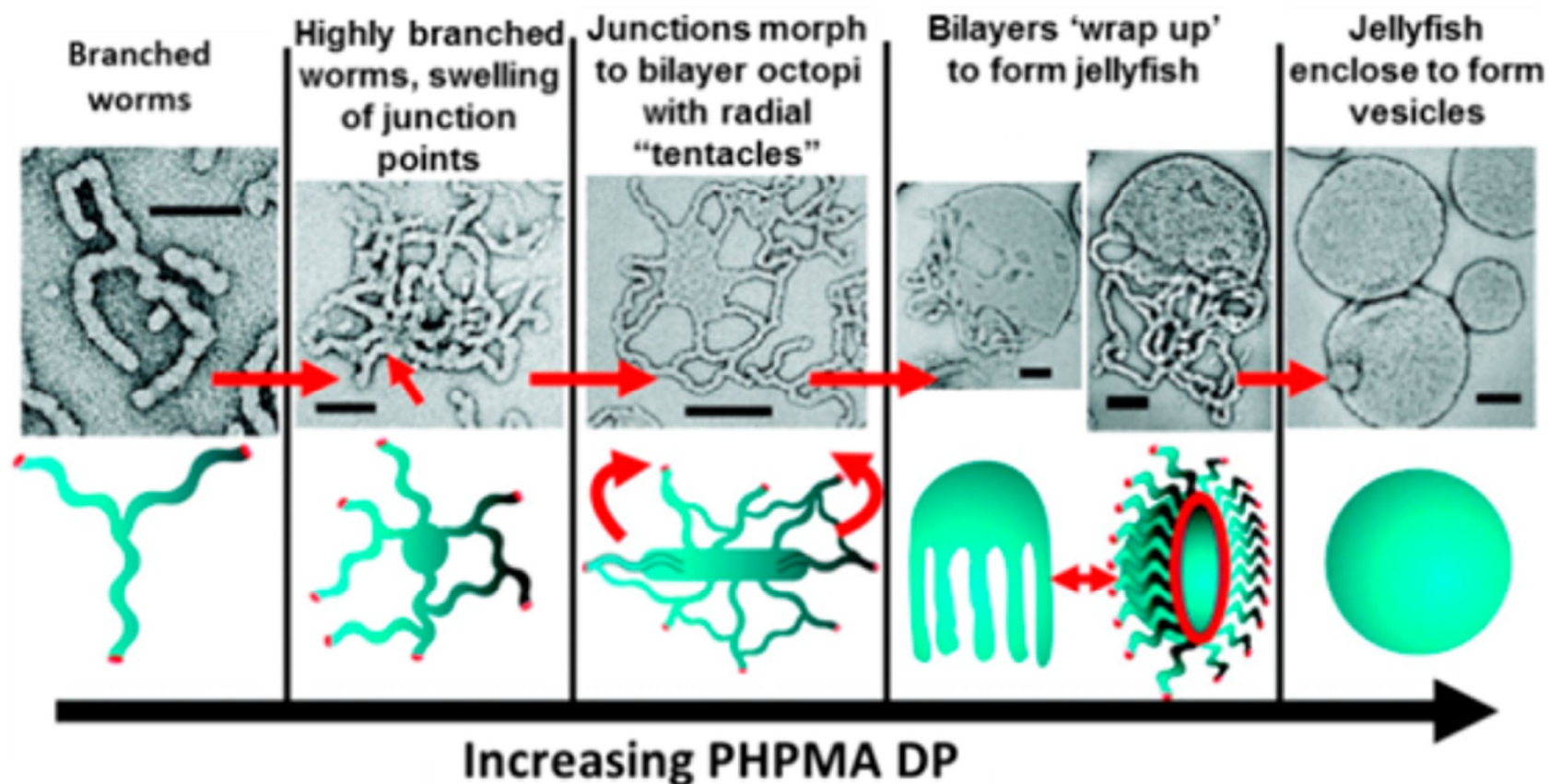
Production of Nanoobjects

Polymerization-Induced Self-Assembly



Production of Nanoobjects

Polymerization-Induced Self-Assembly

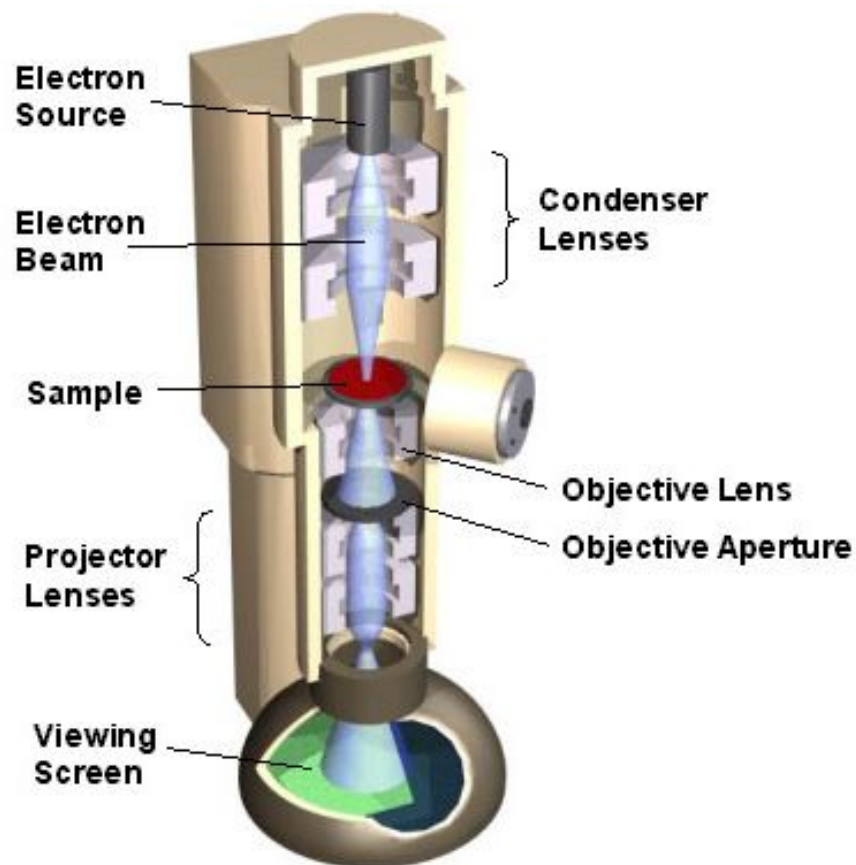


(Purely) Synthetic Block Copolymers
- Materials from Solution Self-Assembly -
Characterization

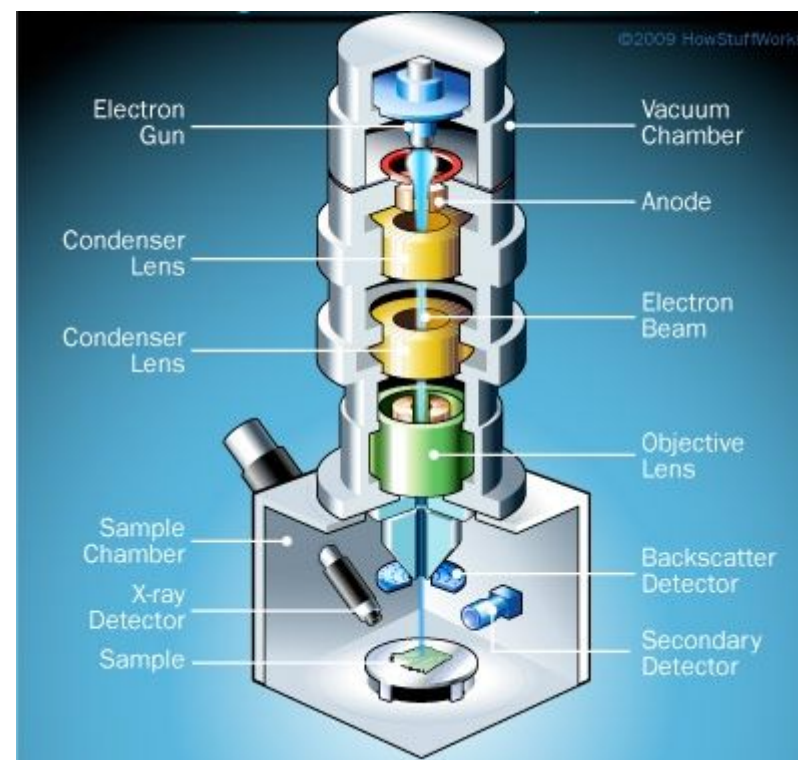
Characterizing the Nanoobjects in Solution

Electron Microscopy

Transmission



Scanning



Characterizing the Nanoobjects in Solution

Electron Microscopy

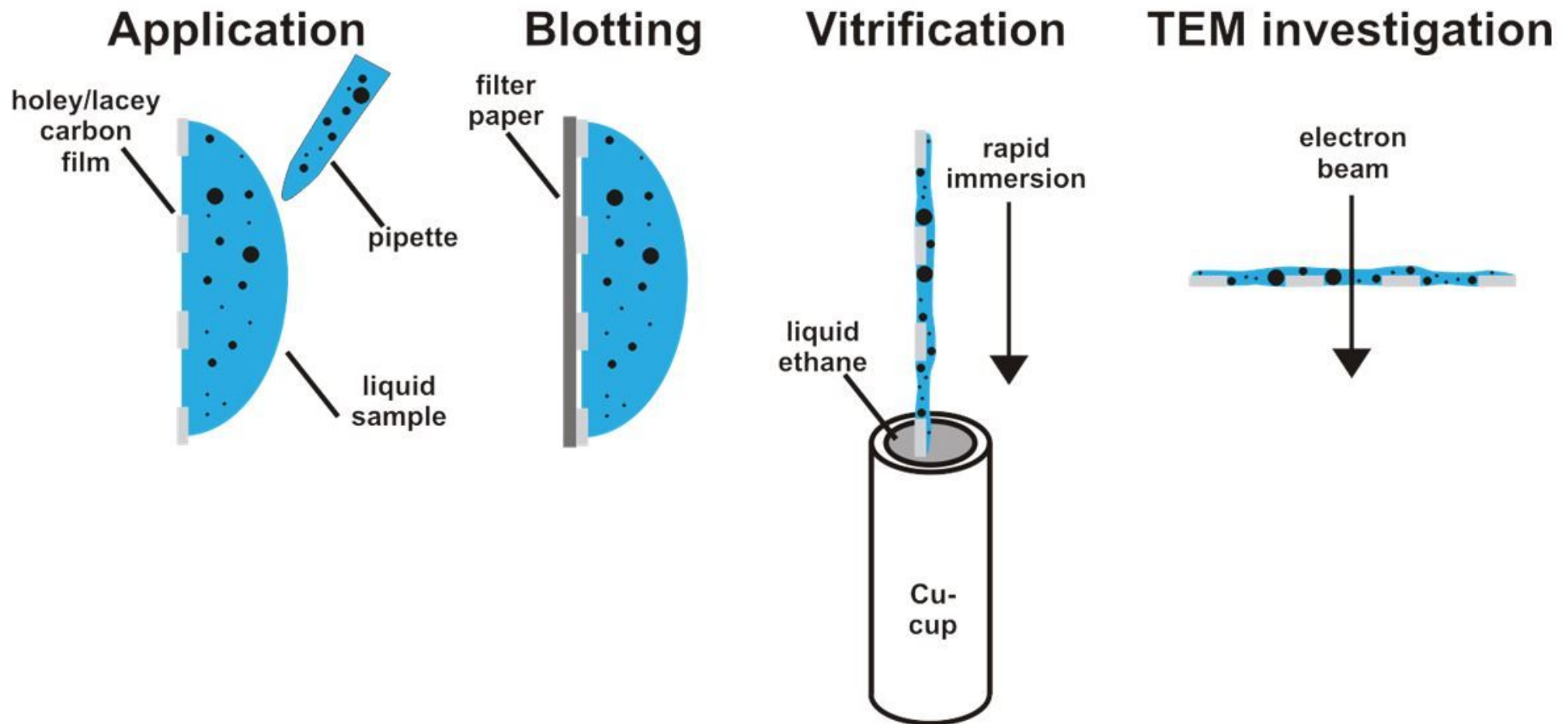
However:

- Some structures are stable *only* in their native environment (concentration, temperature, pH...).
- Soft materials are usually problematic.

Characterizing the Nanoobjects in Solution

Electron Microscopy

Cryogenic Transmission EM (cryo-TEM)

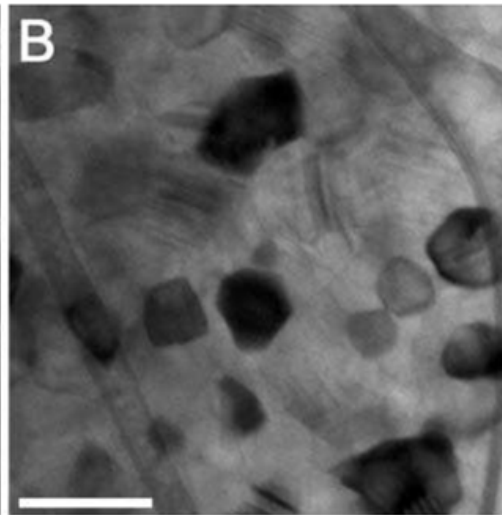
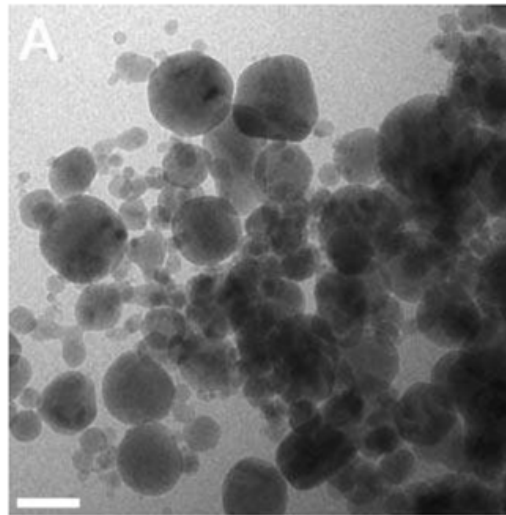


Characterizing the Nanoobjects in Solution

Electron Microscopy

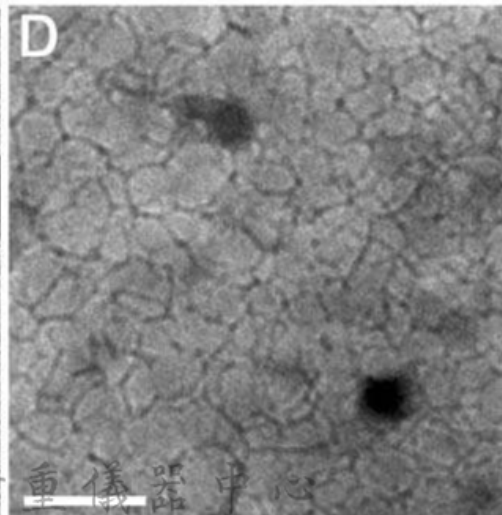
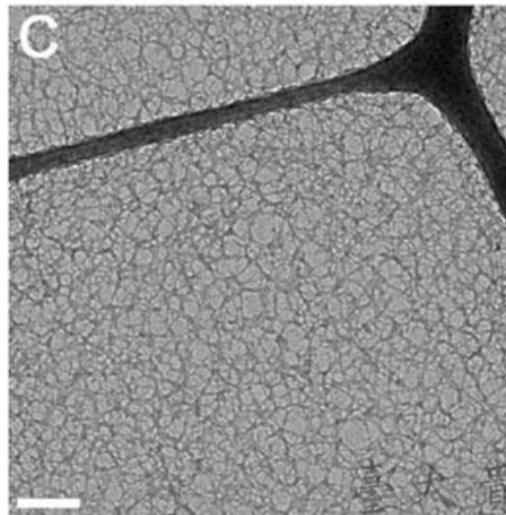
Cryogenic Transmission EM (cryo-TEM)

Large Ice crystal



Hexagonal Ice crystal

Porous dry films of
vitrified water



Scale bar: 200 nm

Characterizing the Nanoobjects in Solution

Electron Microscopy

Cryogenic Transmission EM (cryo-TEM)

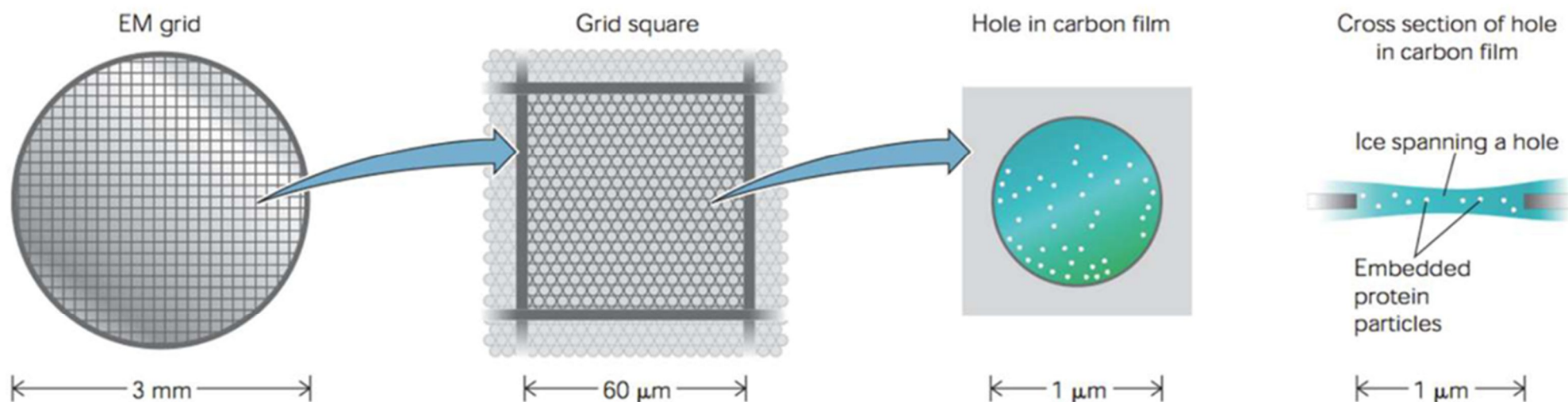


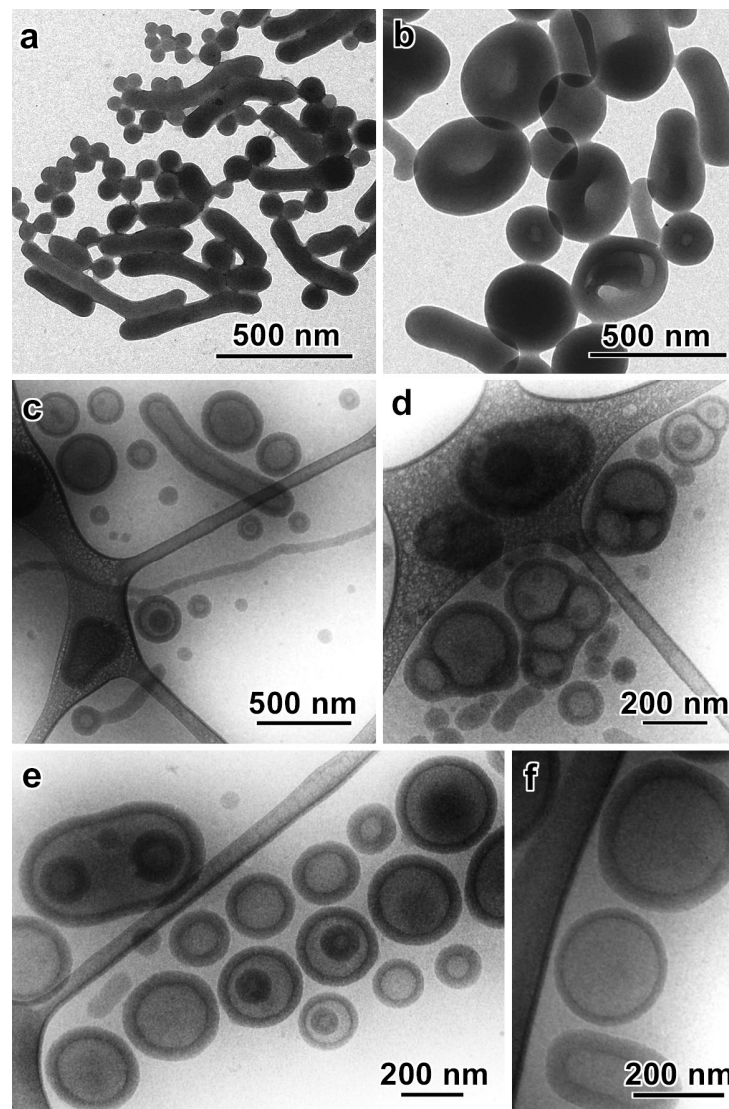
FIGURE 1. The cryo-EM specimen

Adhering to a standard 3-mm electron microscope grid is a carbon film of ~ 500 Å thickness perforated with holes 1–2 μm in diameter. The carbon film supports a 1,000-Å layer of buffer, in which the particles of interest are embedded. This layer is rapidly frozen in liquid ethane to form vitreous ice. The specimen is maintained continuously below -160°C during storage and also during imaging in the electron microscope to prevent the formation of ice crystals.

Characterizing the Nanoobjects in Solution

Electron Microscopy

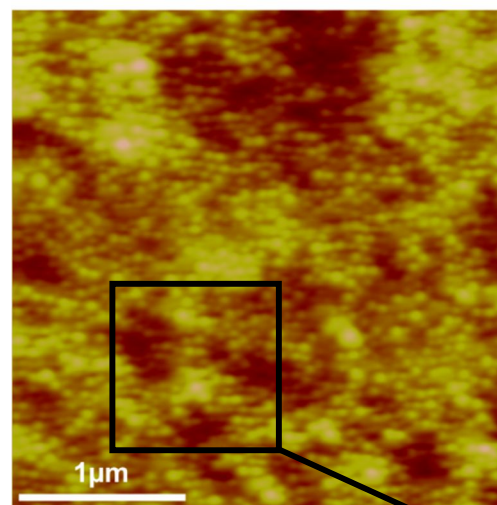
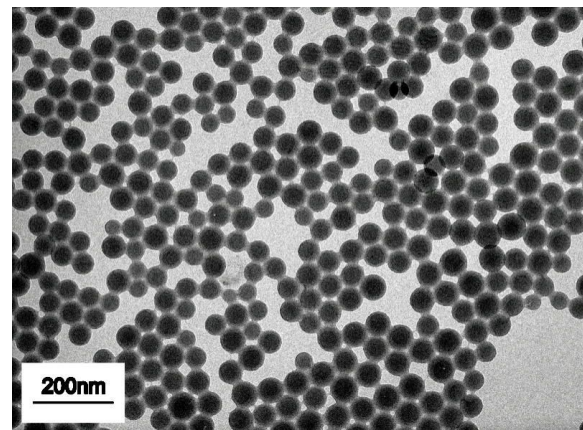
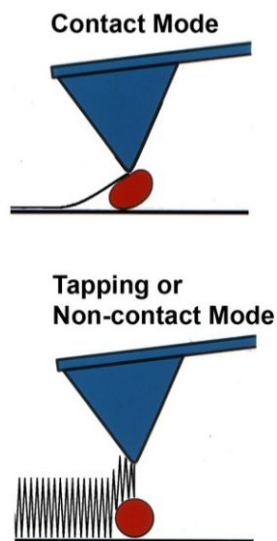
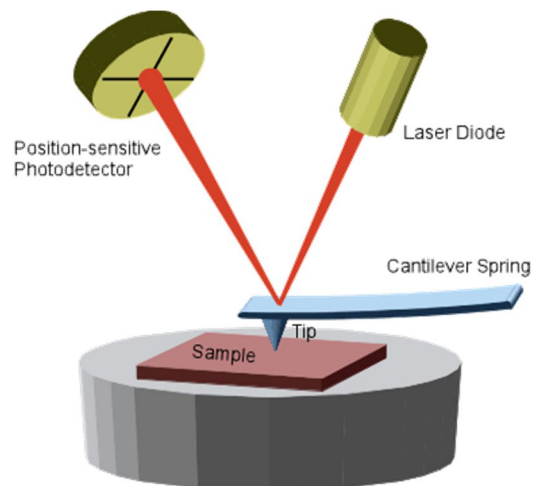
Cryogenic Transmission EM (cryo-TEM)



Delaittre et al.
Chem. Commun. **2009**, 2887

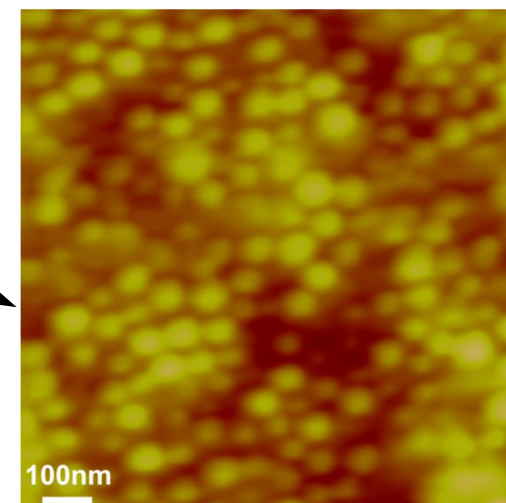
Characterizing the Nanoobjects in Solution

Atomic Force Microscopy (AFM)



Dilution 1%
On mica
Drying

X 3

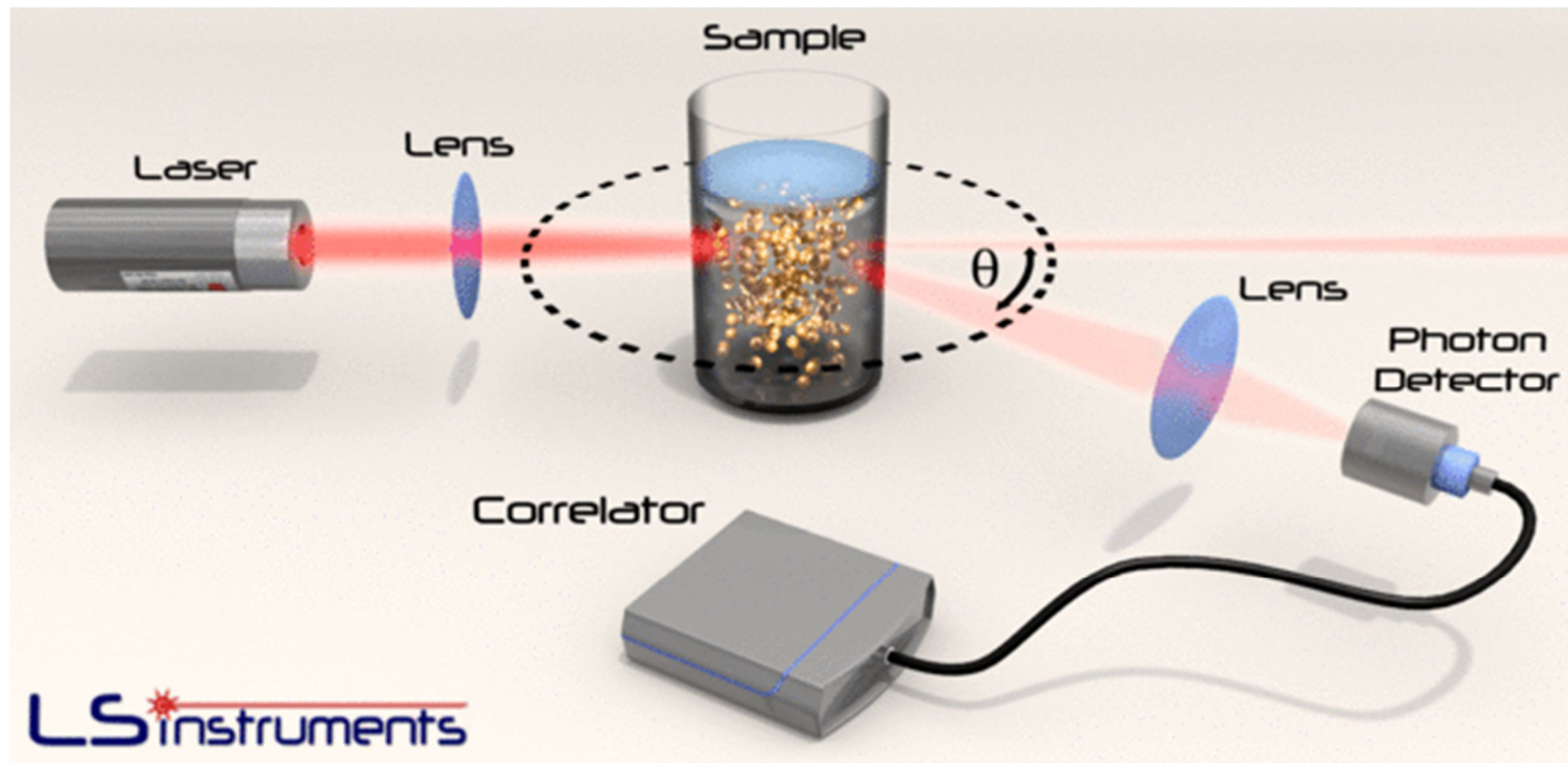


Characterizing the Nanoobjects in Solution

Light Scattering

Dynamic (DLS)

also called photon correlation spectroscopy or quasi-elastic light scattering

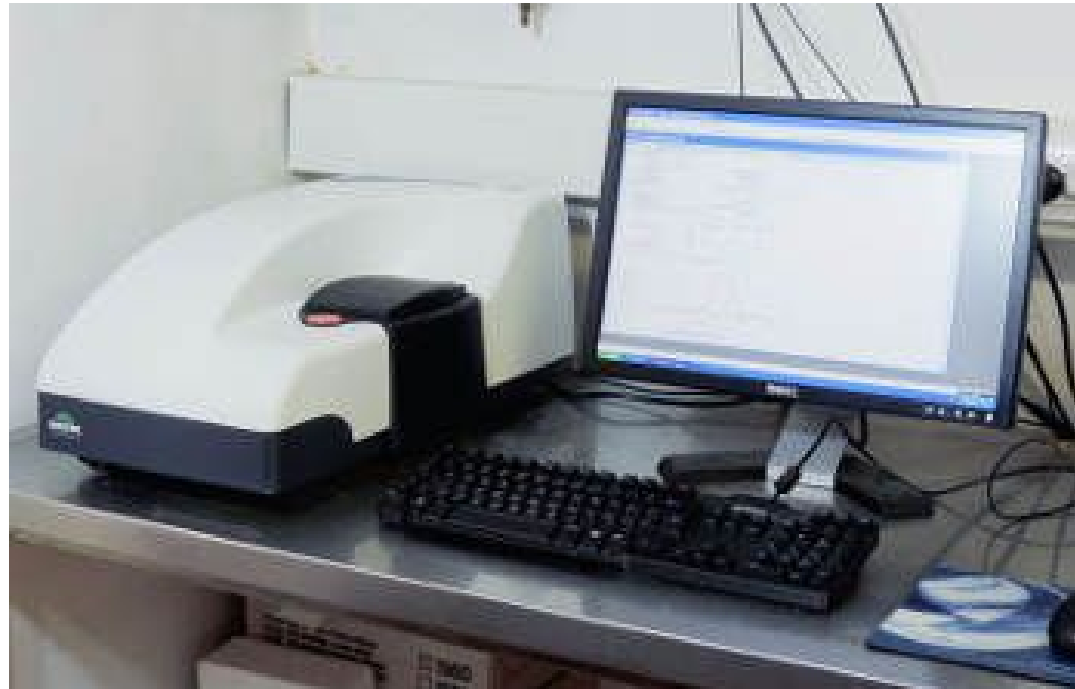


Characterizing the Nanoobjects in Solution

Light Scattering

Dynamic (DLS)

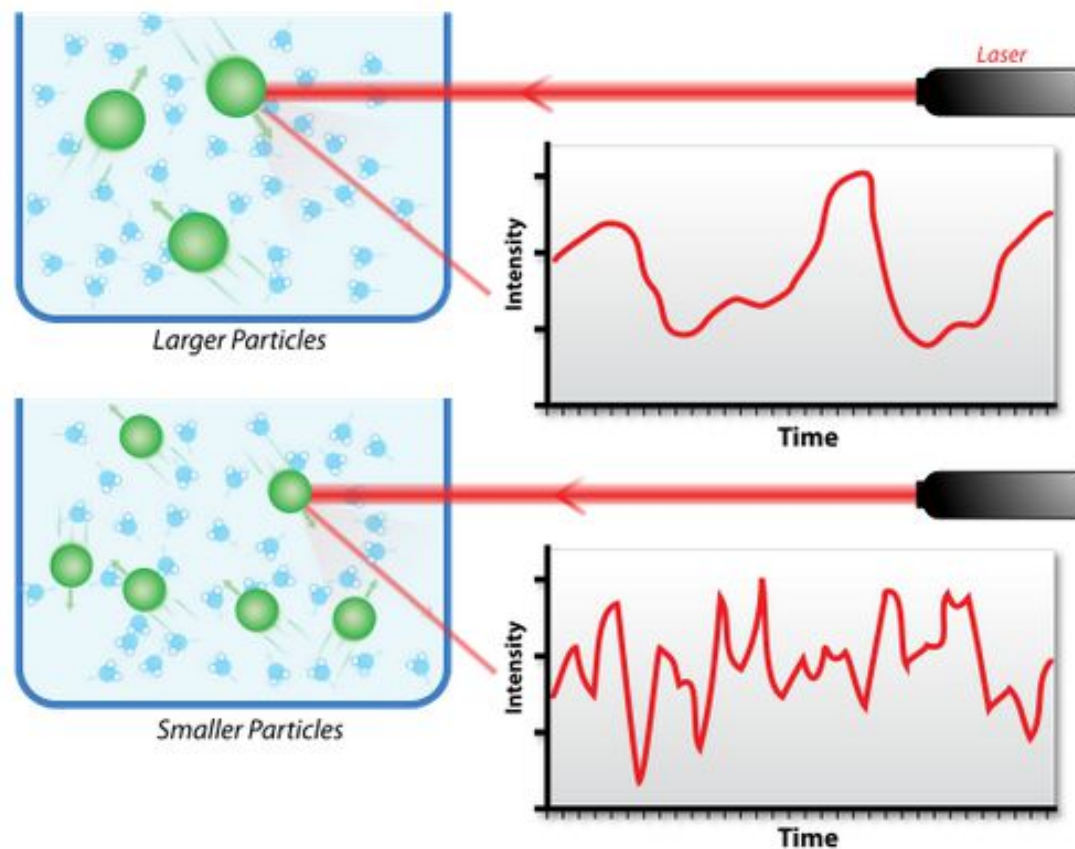
also called photon correlation spectroscopy or quasi-elastic light scattering



Characterizing the Nanoobjects in Solution

Light Scattering

Dynamic (DLS)

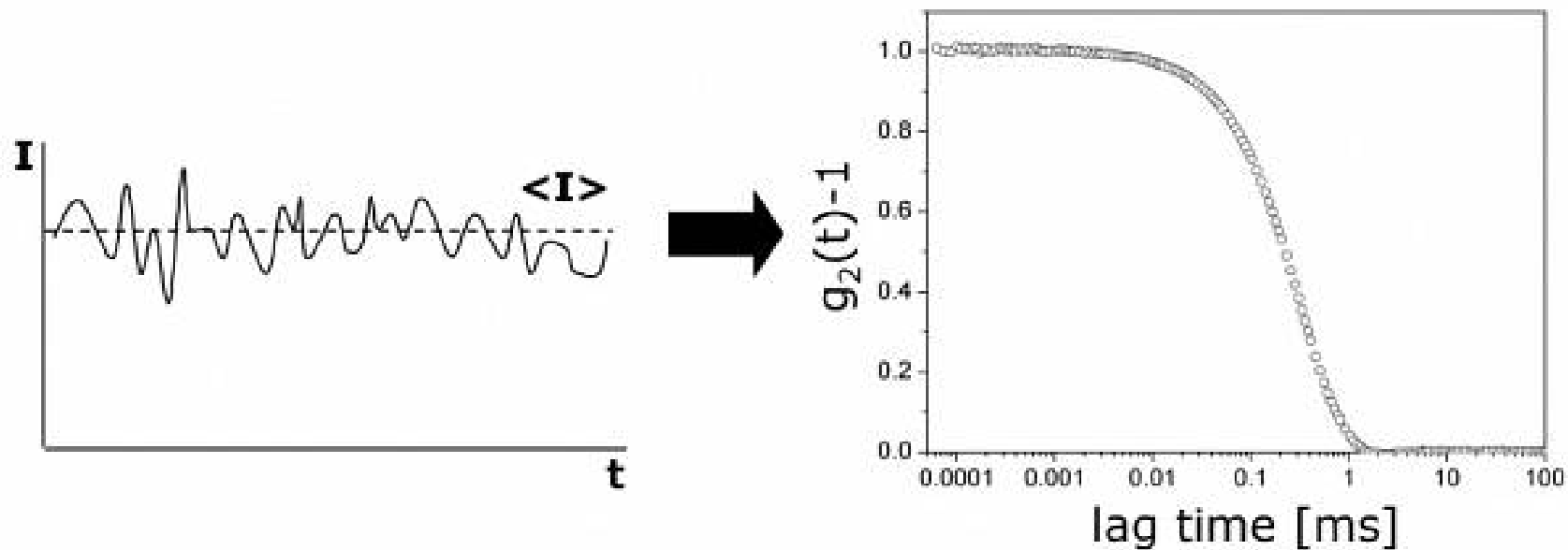


https://en.wikipedia.org/wiki/Dynamic_light_scattering

Characterizing the Nanoobjects in Solution

Light Scattering

Dynamic (DLS)

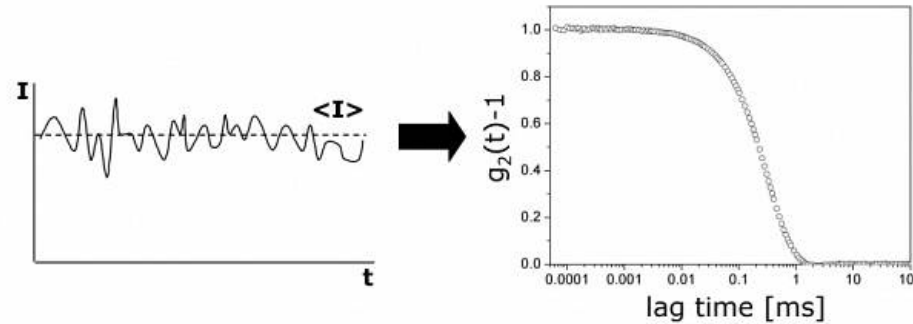


$$g^2(q; \tau) = \frac{\langle I(t)I(t + \tau) \rangle}{\langle I(t) \rangle^2} = 1 + \beta [g^1(q; \tau)]^2$$

Characterizing the Nanoobjects in Solution

Light Scattering

Dynamic (DLS)



$$g^1(q; \tau) = \exp(-\Gamma \tau)$$

$$\Gamma = q^2 D_t \quad \text{with} \quad q = \frac{4\pi n_0}{\lambda} \sin\left(\frac{\theta}{2}\right)$$

$$D = \frac{k_B T}{6\pi \eta r}$$

Stokes-Einstein equation

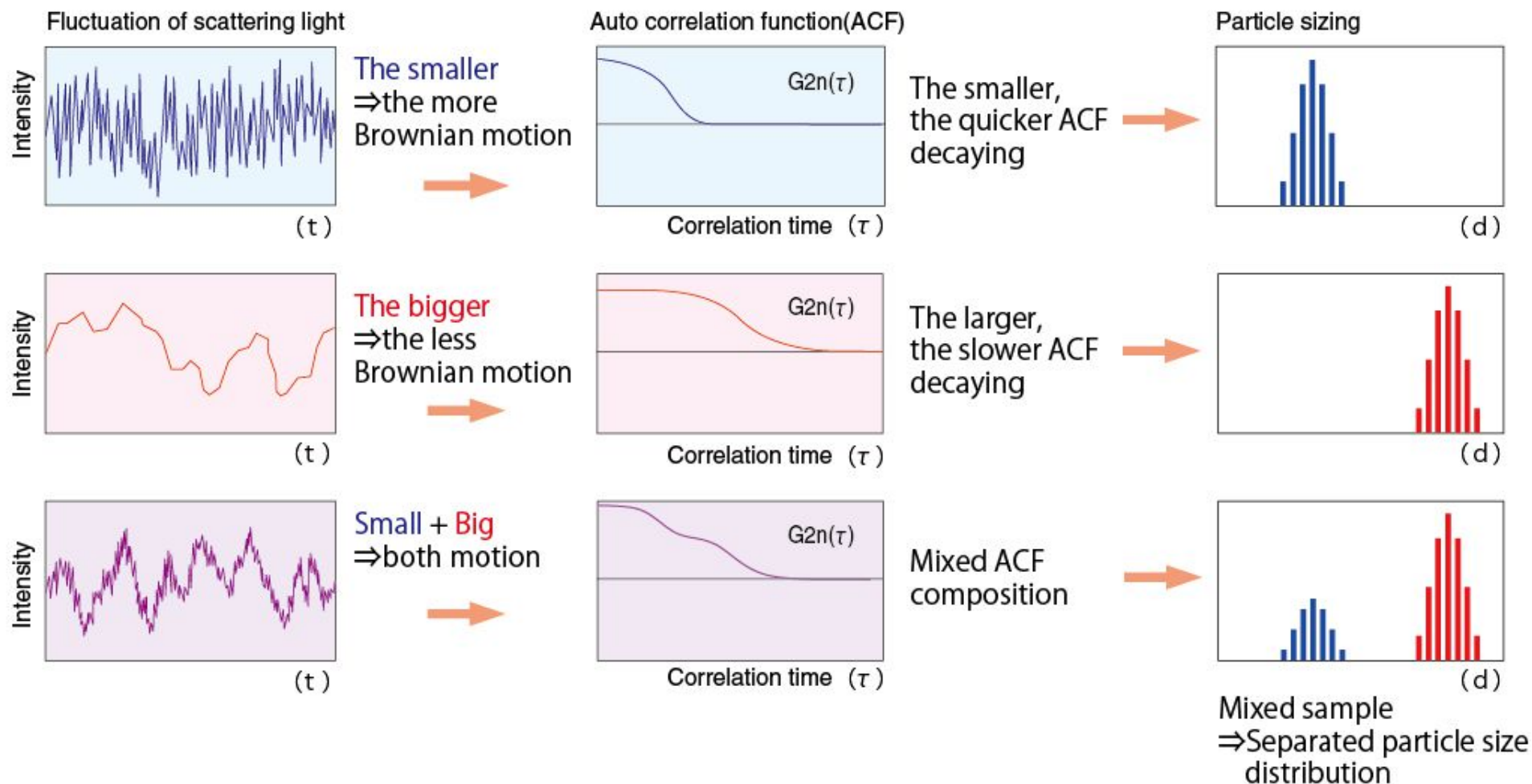
hydrodynamic radius! R_h !!!

= radius of an equivalent hard sphere diffusing at the same rate as the molecule under observation

Characterizing the Nanoobjects in Solution

Light Scattering

Dynamic (DLS)

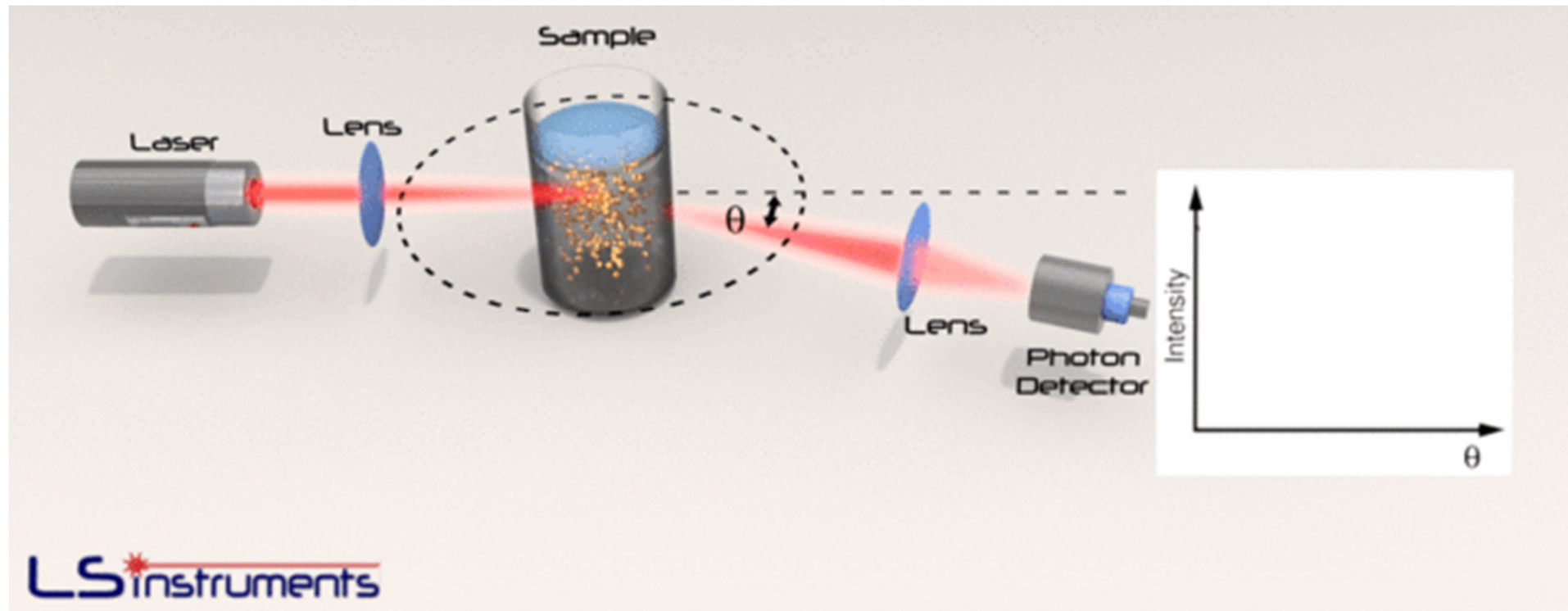


<https://www.otsukael.com/product/detail/productid/23>

Characterizing the Nanoobjects in Solution

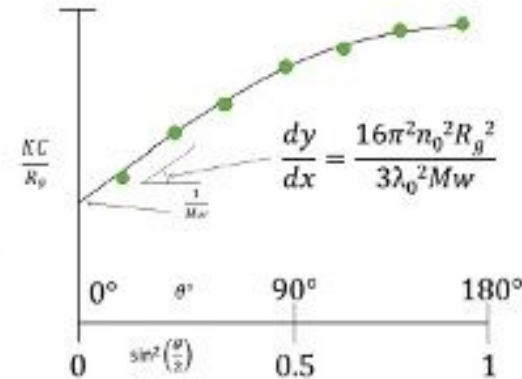
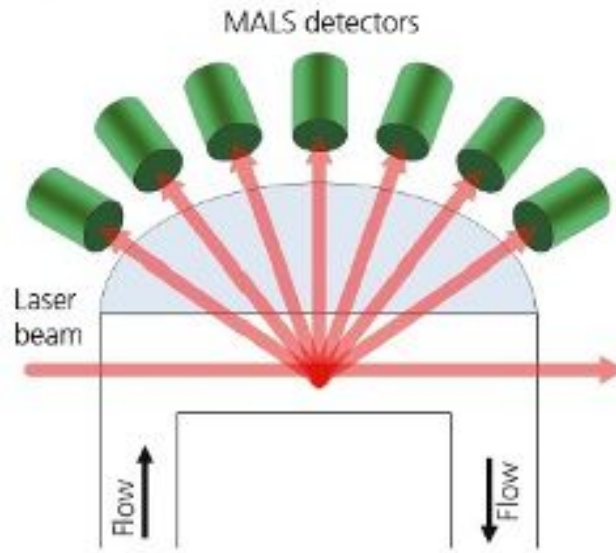
Light Scattering

Static (SLS)



Characterizing the Nanoobjects in Solution

Light Scattering Static (SLS)



Zimm equation

Intrinsic values
(n , dn/dc , λ)

Intrinsic values (n , ϑ , λ)

weight-average molar mass

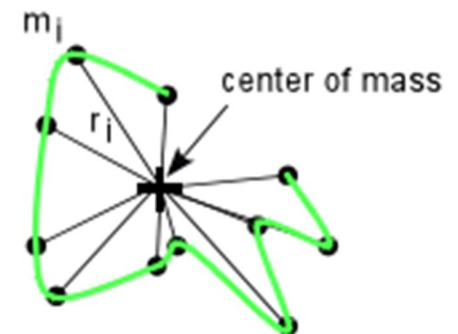
$$\frac{Kc}{\Delta R(\theta, c)} = \frac{1}{M_w} \left(1 + \frac{q^2 R_g^2}{3} + O(q^4) \right) (1 + 2M_w A_2 c + O(c^2))$$

$$R(\theta) = \frac{I_A(\theta)n_0^2}{I_T(\theta)n_T^2} \frac{R_T}{N(\theta)}$$

radius of gyration

= mass weighted average distance from the core of a molecule to each mass element in the molecule

$$x_{\text{rms}} = \sqrt{\frac{1}{n} (x_1^2 + x_2^2 + \dots + x_n^2)}.$$



Characterizing the Nanoobjects in Solution

Light Scattering

The ratio $\rho = R_g/R_h$ (shape factor) can provide information on the shape of nanoobjects.

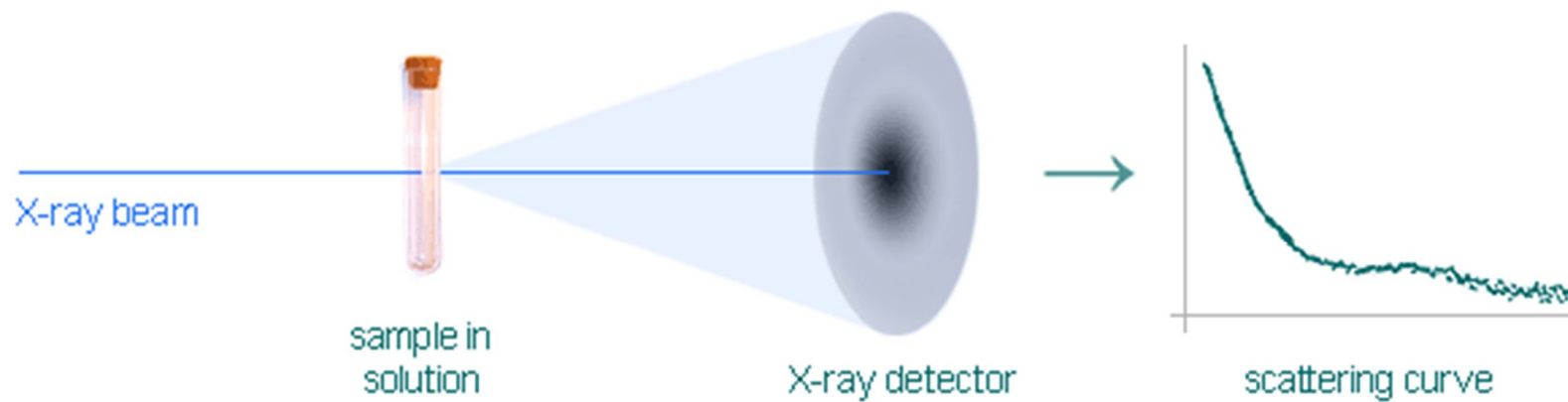
$\rho = 0.775$ globule/sphere

$\rho = 0.816$ coil

$\rho = 1.732$ rod

Characterizing the Nanoobjects in Solution

Small-Angle X-Ray Scattering



Characterizing the Nanoobjects in Solution

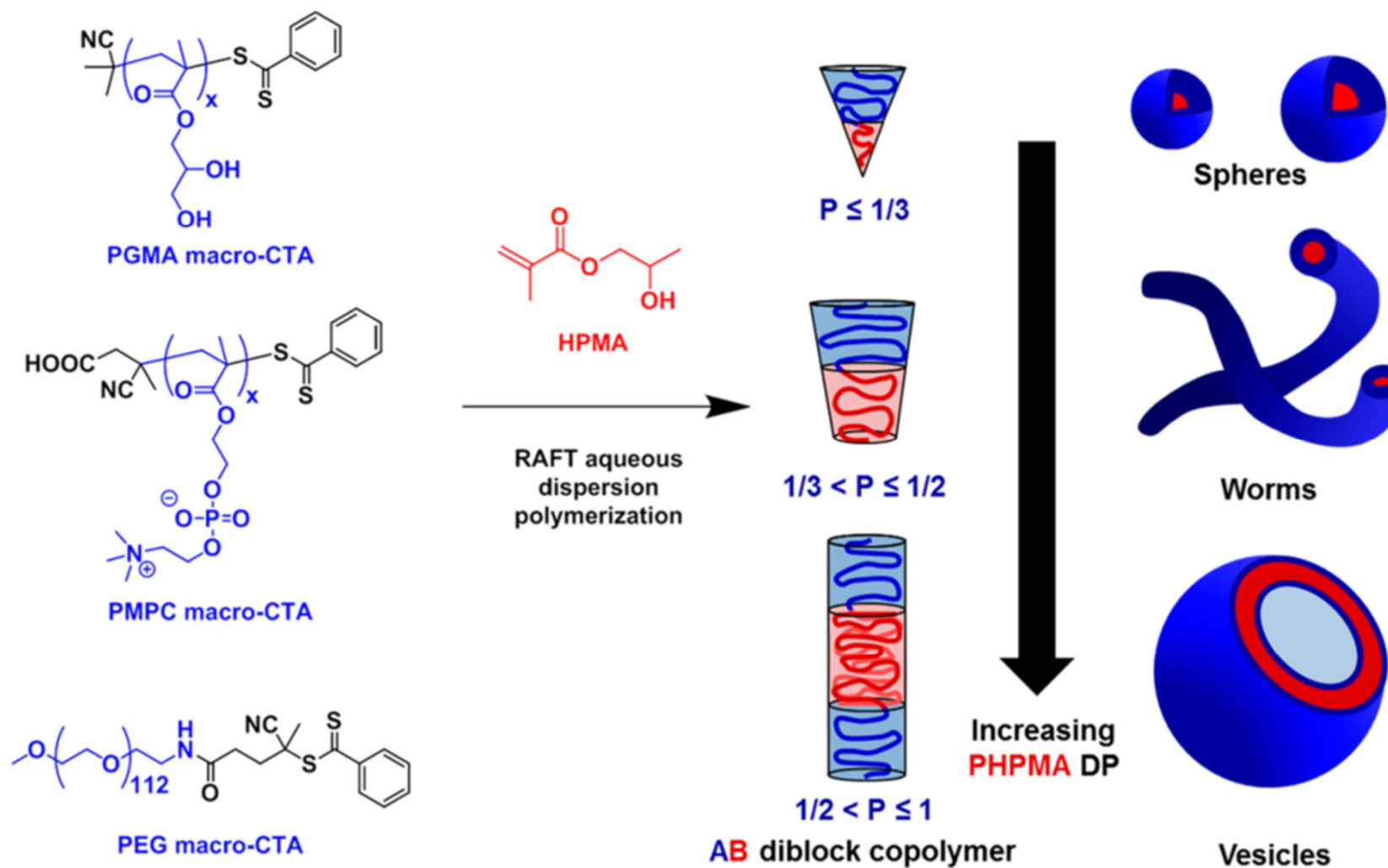
Small-Angle X-Ray Scattering



biosaxs.com

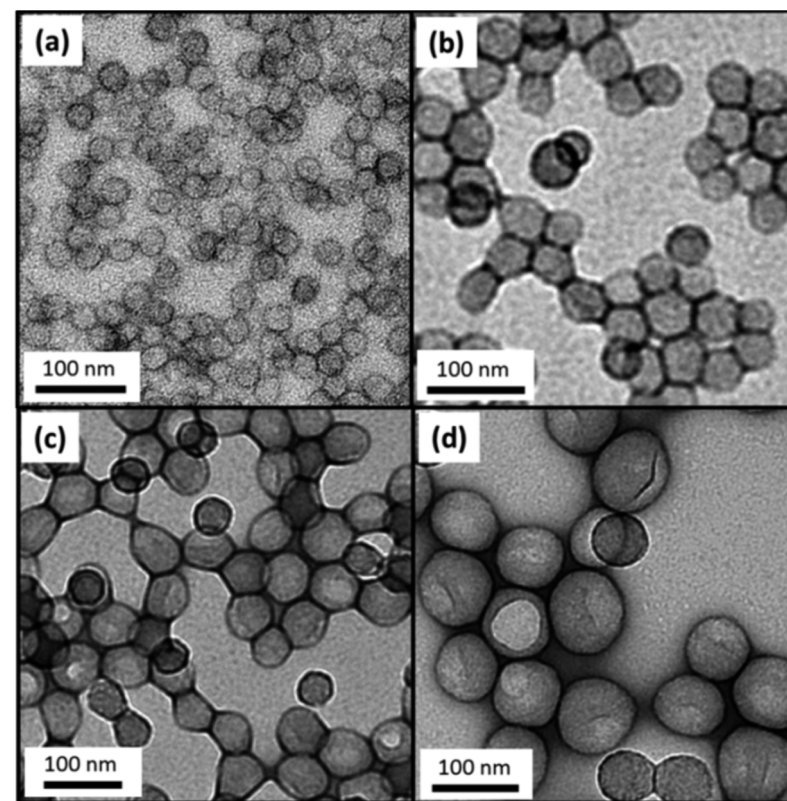
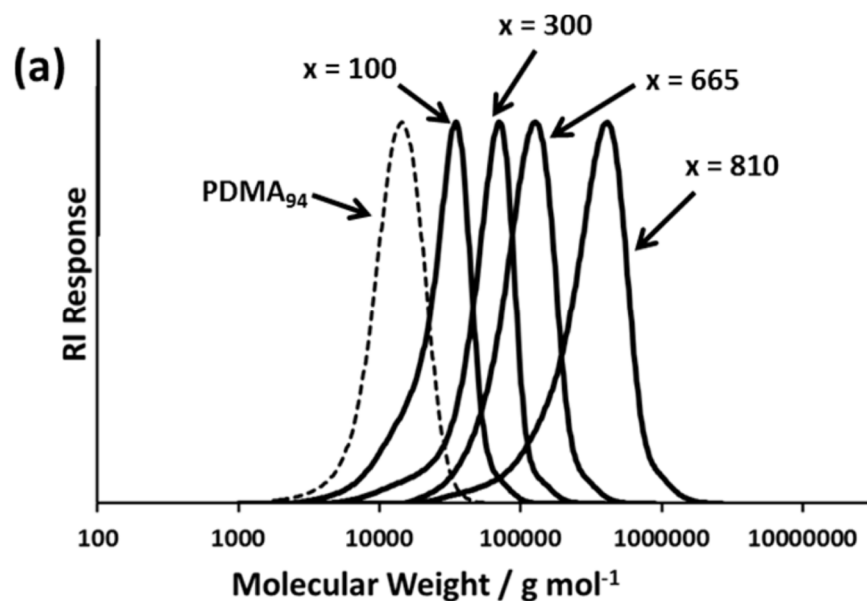
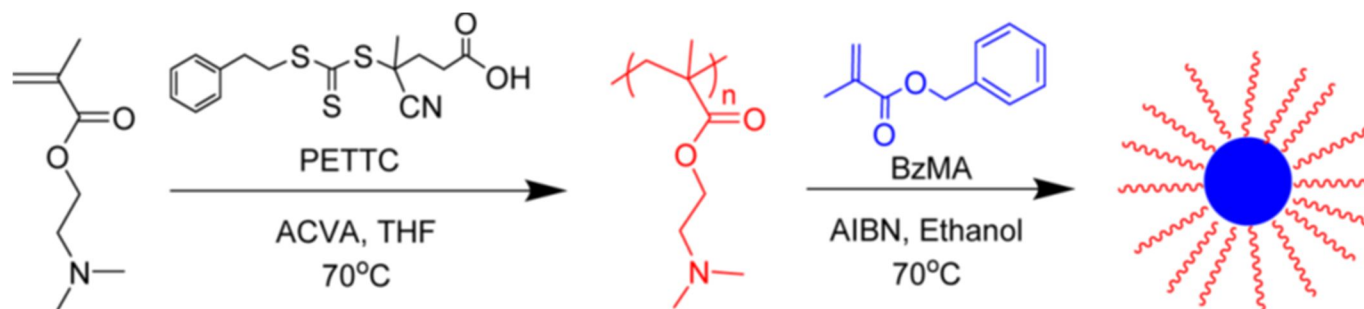
Characterizing the Nanoobjects in Solution

Small-Angle X-Ray Scattering



Characterizing the Nanoobjects in Solution

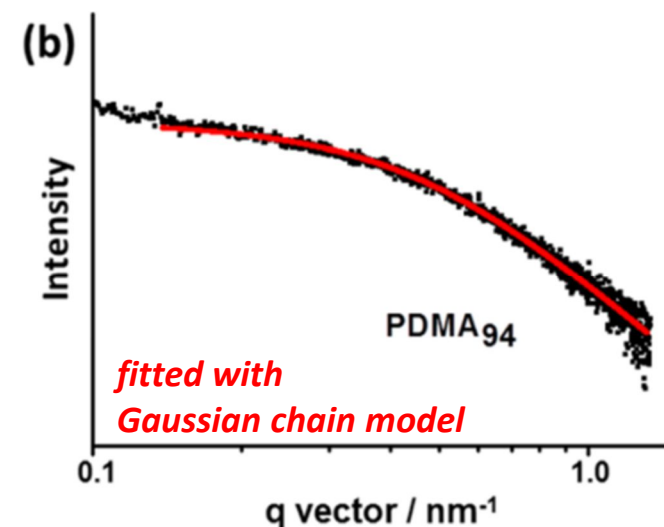
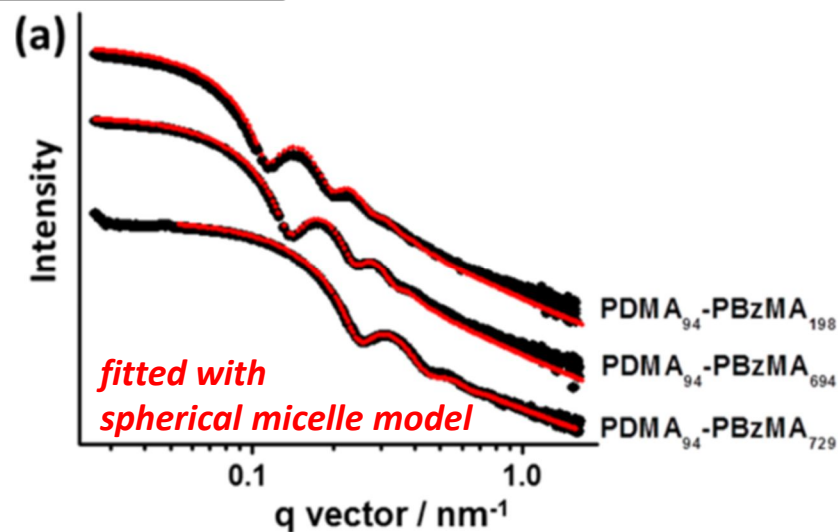
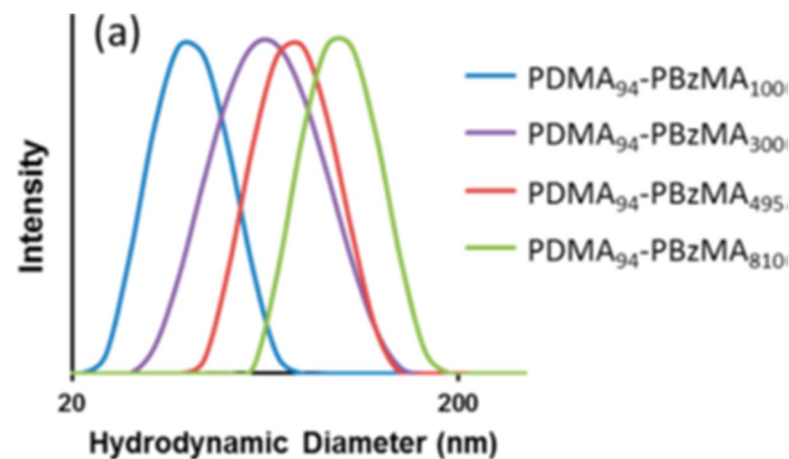
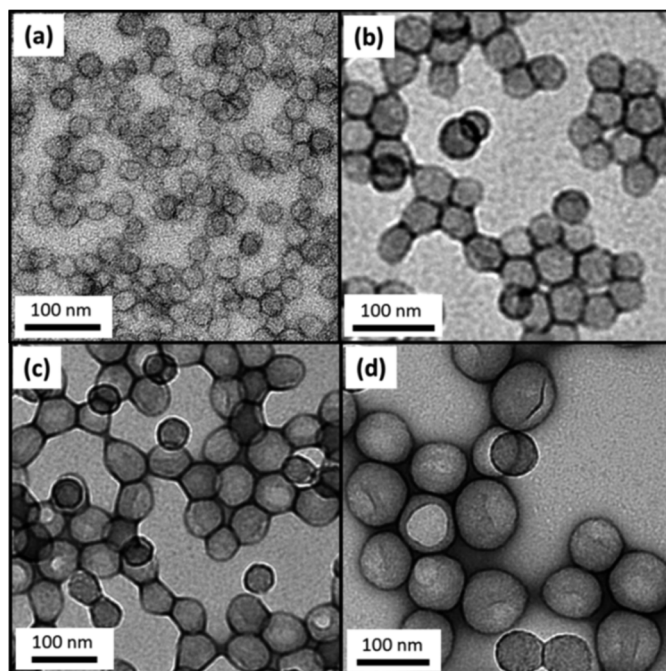
Small-Angle X-Ray Scattering



Macromolecules 2016, 49, 172

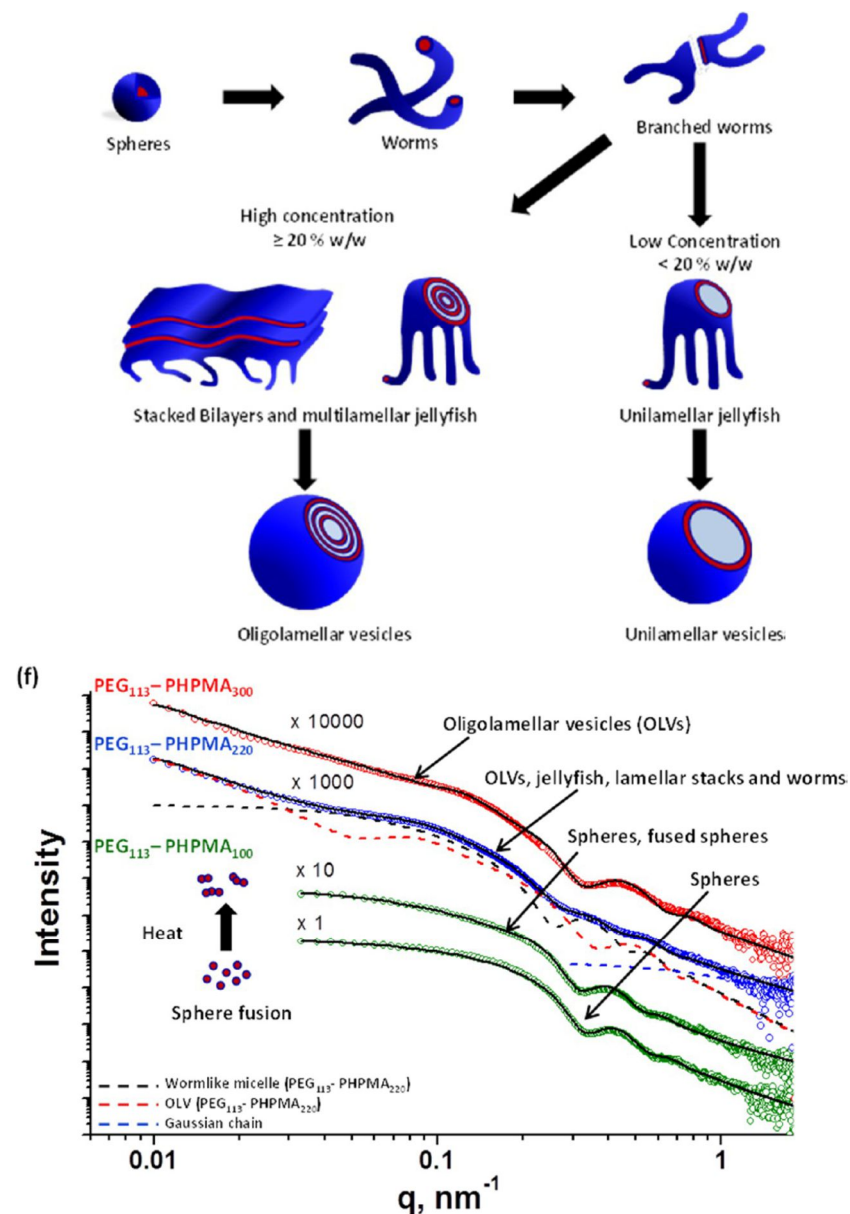
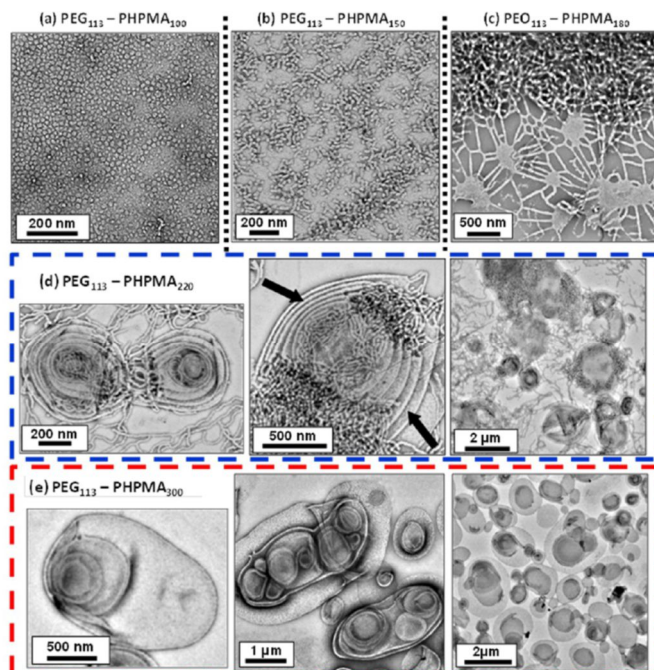
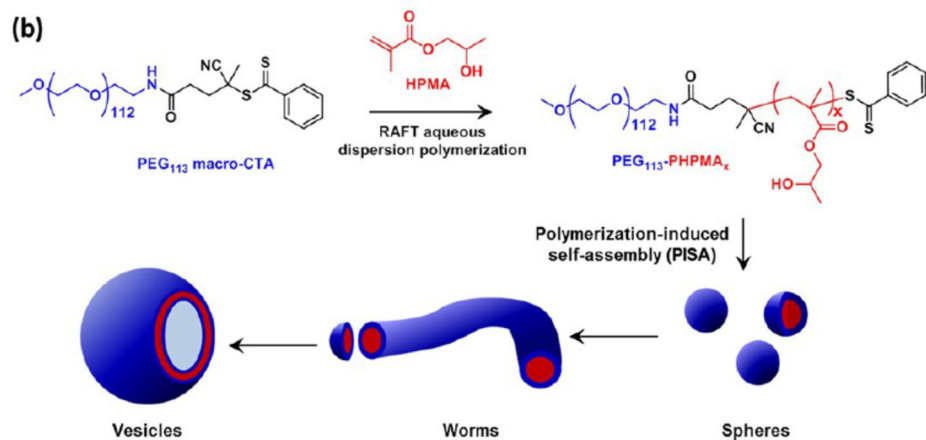
Characterizing the Nanoobjects in Solution

Small-Angle X-Ray Scattering



Characterizing the Nanoobjects in Solution

Small-Angle X-Ray Scattering



Small-Angle Neutron Scattering

Reminder

The important in scattering-based characterization methods is that there is a **contrast between the particle (or pore) and the surrounding material in the sample.**

The type of contrast varies depending on the type of radiation used.

Light scattering	transmittance contrast
SAXS	electron density contrast
SANS	neutron scattering cross-section

A cross-section is a value expressing the likelihood for a given target object to interact with an incident particle

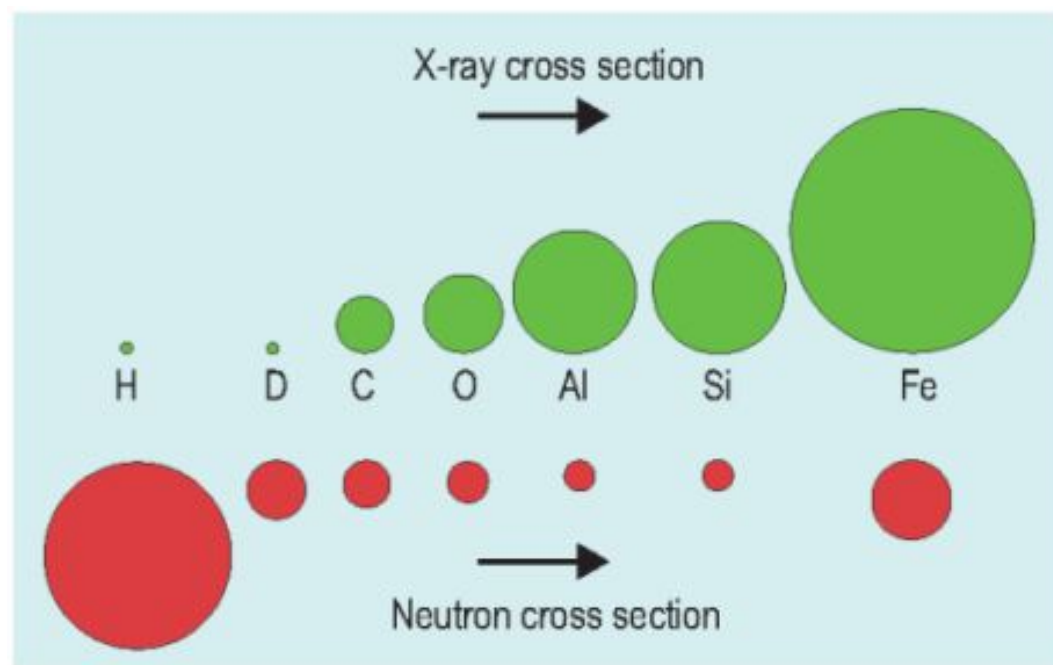
Characterizing the Nanoobjects in Solution

Small-Angle Neutron Scattering

X-rays interact with the **electronic cloud**... **neutrons** with the **nucleus**.

Advantages of SANS over SAXS

- sensitivity to light elements
- possibility of isotope labelling



Less than 50 SANS-capable facilities in the world (2025)

Characterizing the Nanoobjects in Solution

Small-Angle Neutron Scattering

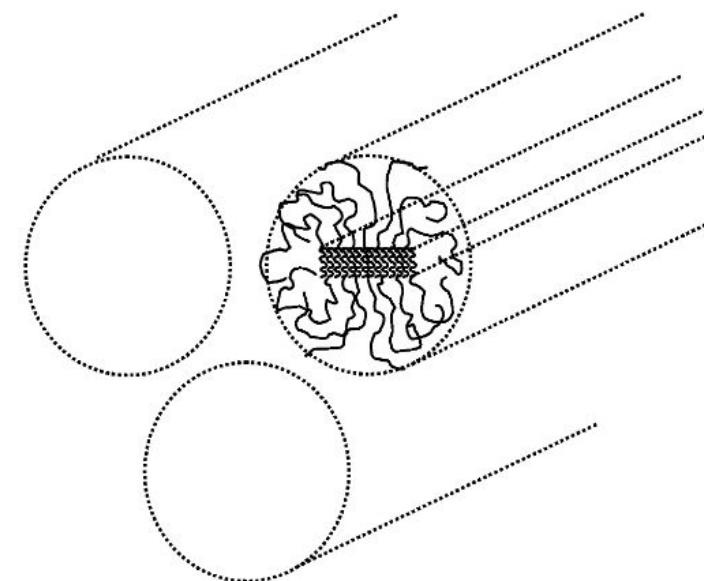
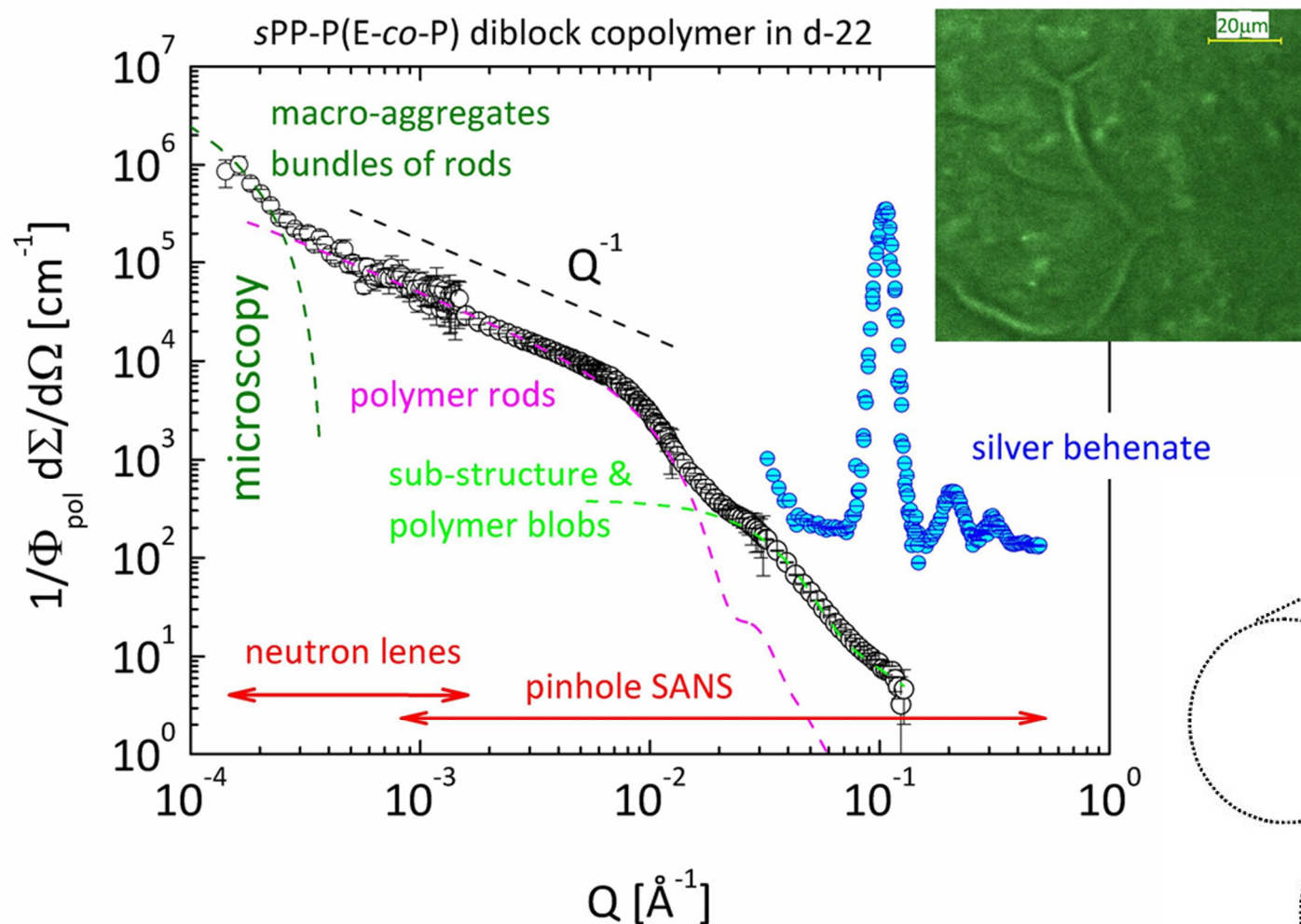
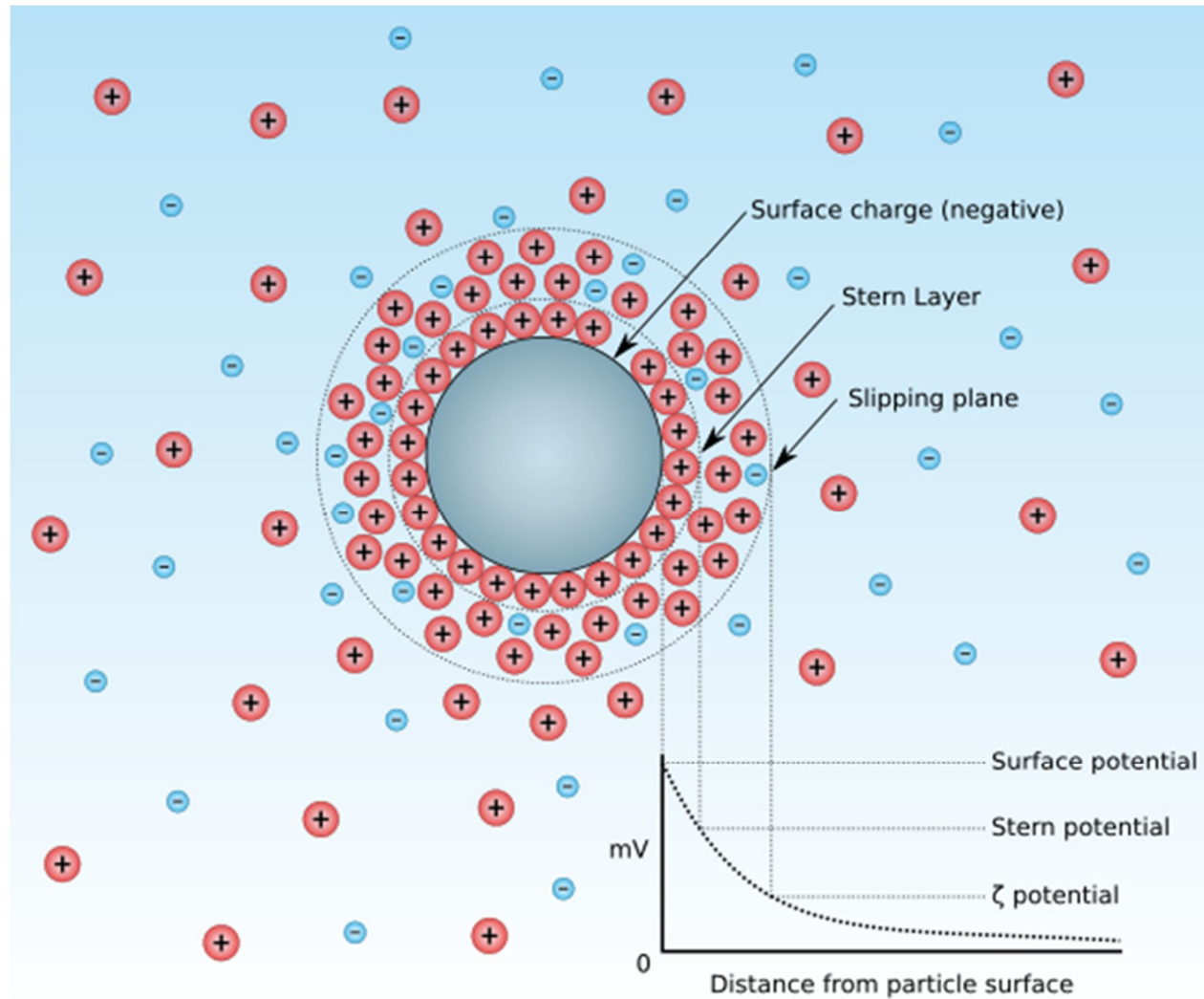


Figure 13. Schematic representation of a possible morphology of the aggregates formed by sPP-P(E-co-P) diblock copolymers in decane: bundle of polymeric rods made by crystalline/amorphous sequences.

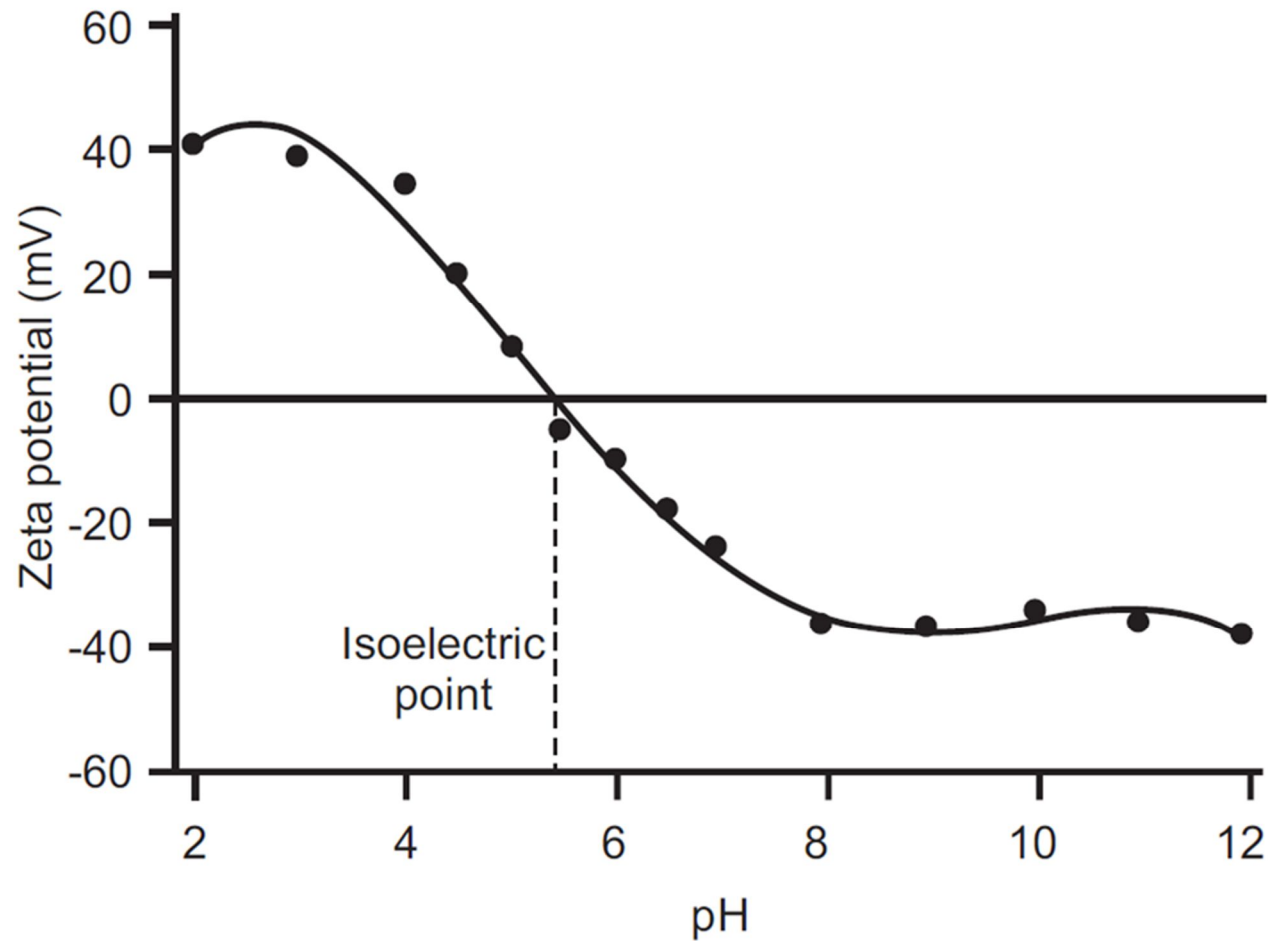
Characterizing the Nanoobjects in Solution

Zeta Potential



Characterizing the Nanoobjects in Solution

Zeta Potential



Characterizing the Nanoobjects in Solution

Zeta Potential

Henry's equation (electrophoretic mobility)

$$\mu_E = \frac{2 \varepsilon \zeta f(\kappa a)}{3\eta}$$

ε dielectric constant

η viscosity

ζ zeta potential

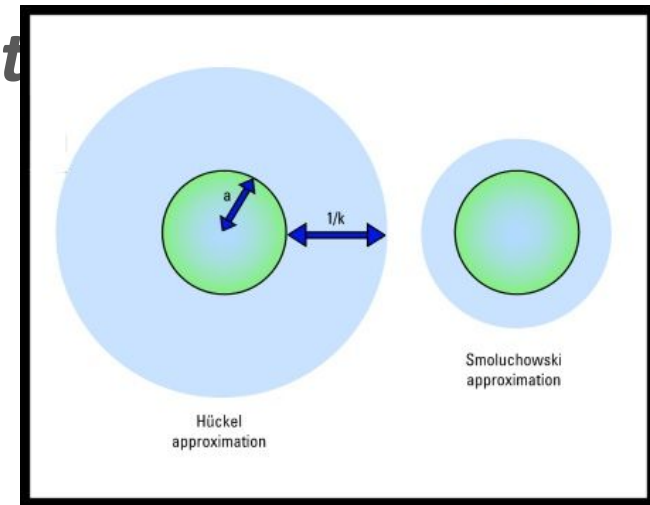
$f(\kappa a)$ Henry's function

approximations **Smoluchowski** 1.5 ($D > 200$ nm, > 1 mM salt)

Hückel 1.0 small particles in low ε media

aqueous media

non-aqueous media



In practice

$$v_0 = \frac{\mu_E}{E_0}$$

v_0 velocity

E_0 applied electrical field

Characterizing the Nanoobjects in Solution

Zeta Potential

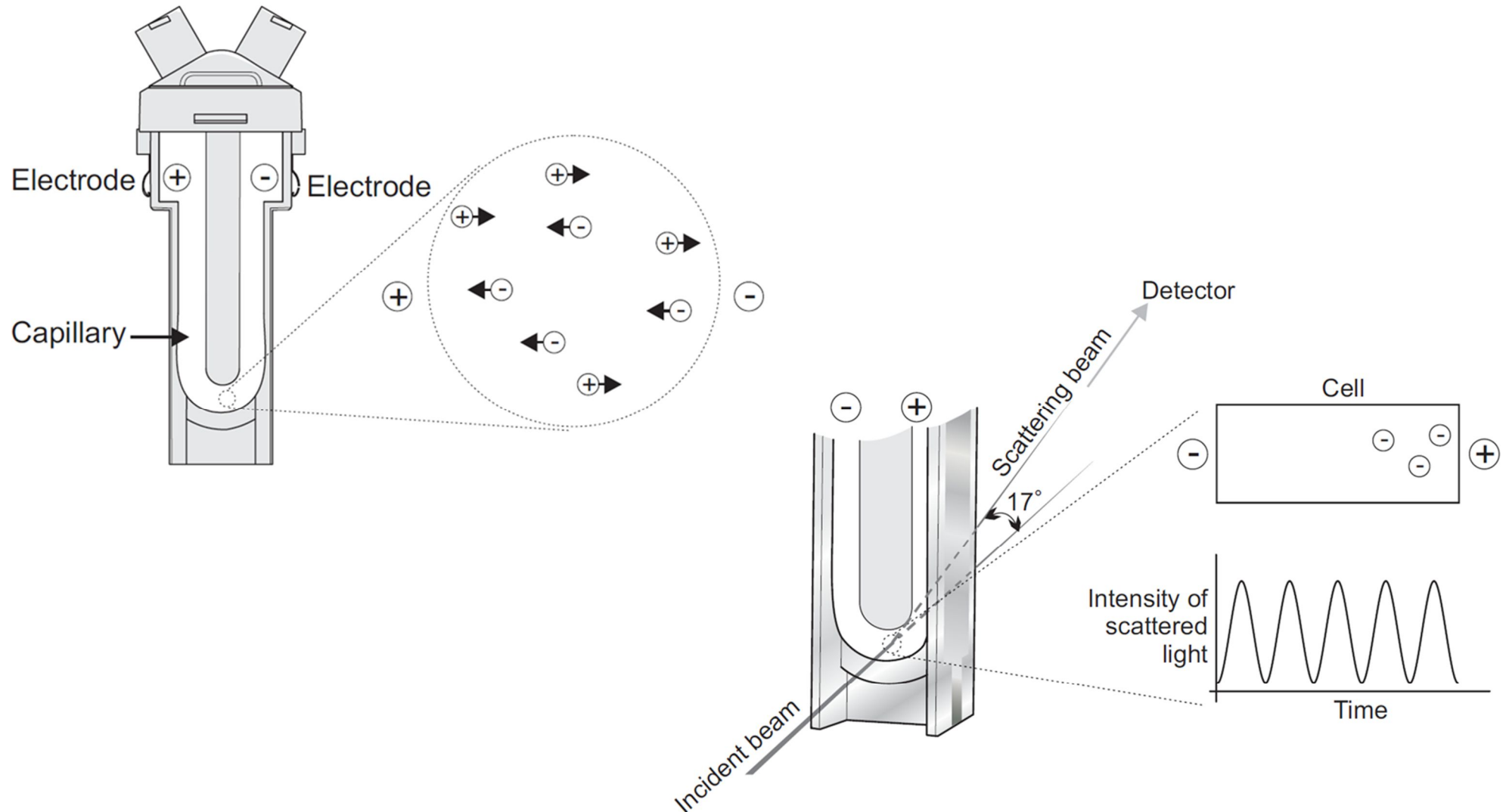
Measuring electrophoretic mobility



Characterizing the Nanoobjects in Solution

Zeta Potential

Measuring electrophoretic mobility



(Purely) Synthetic Block Copolymers
- Materials from Solution Self-Assembly -
Applications

Applications of BCP Solution Self-Assemblies

Block copolymers often **position** themselves at **interfaces**.

They can thus influence the stability of dispersions and therefore act as: emulsifiers, wetting agents, foam stabilizers, flocculants, etc...

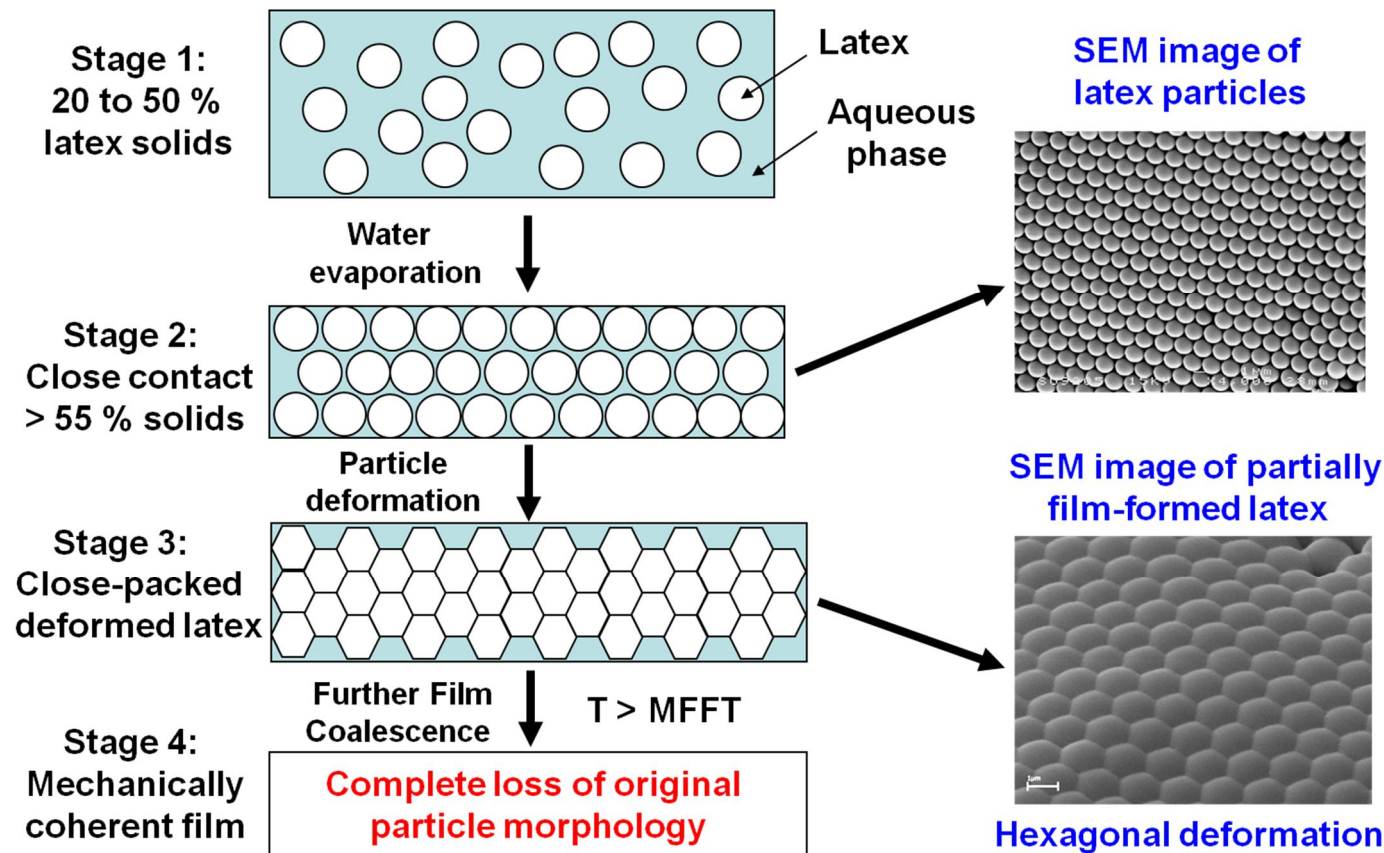
Applications of BCP Solution Self-Assemblies

Particle/Emulsion stabilization

Advantages latex

- Reduced viscosity
- Reduced air pollution as no volatile organic solvents are present

Film formation mechanism



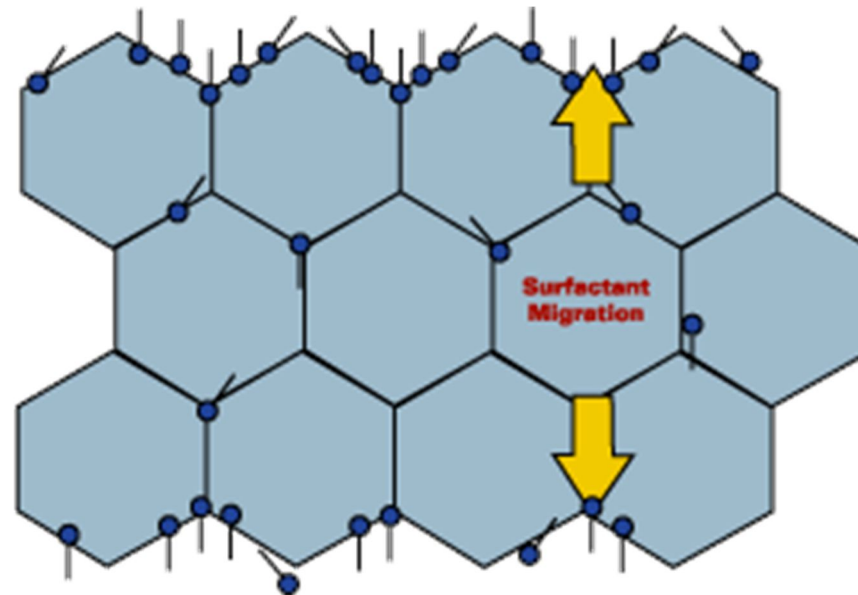
Jones et al., 1995

Particle/Emulsion stabilization

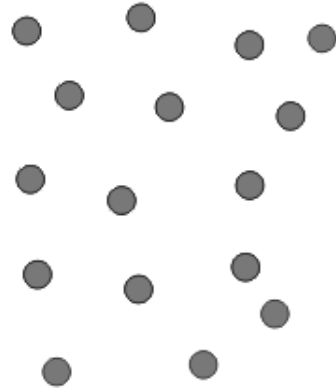
BCP vs. surfactants

Surfactants can cluster and migrate to interfaces across the film during formation, which affects various properties:

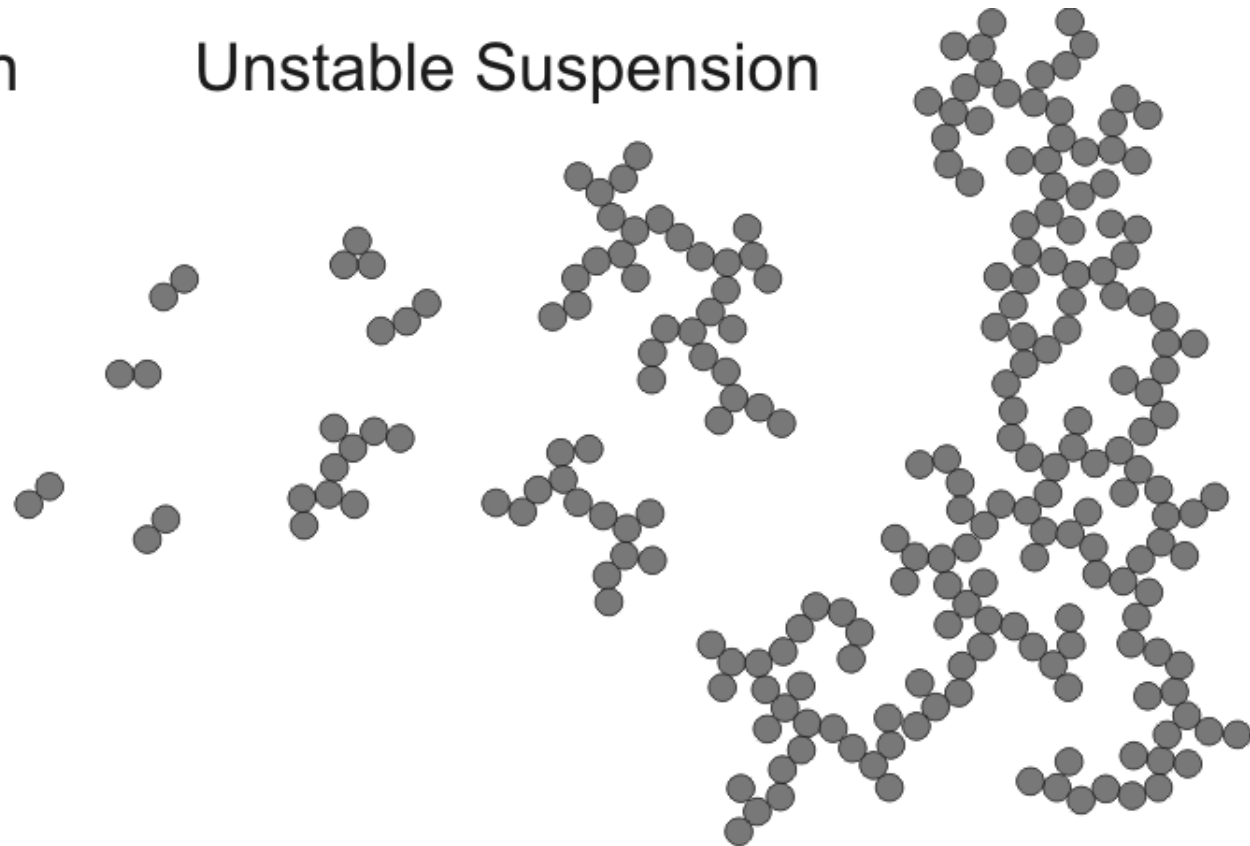
- Resistance (hydrophilic channels)
- Outdoor durability
- Toxicity
- Adhesion

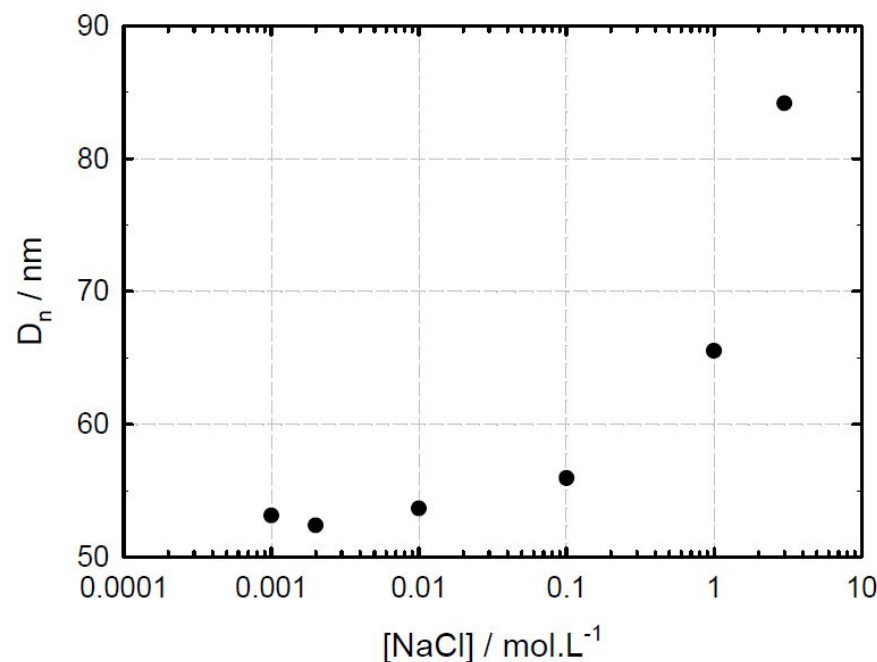
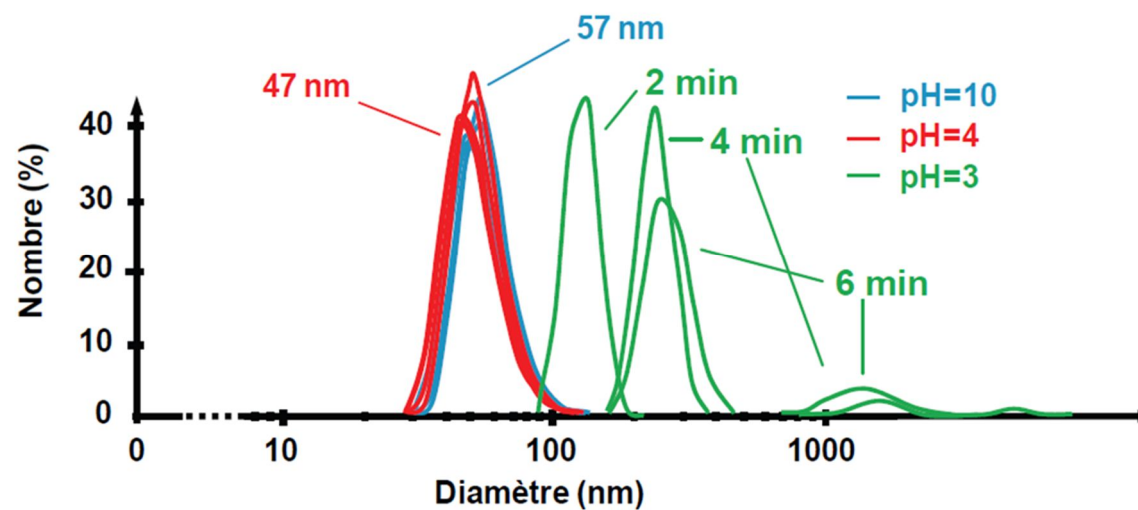
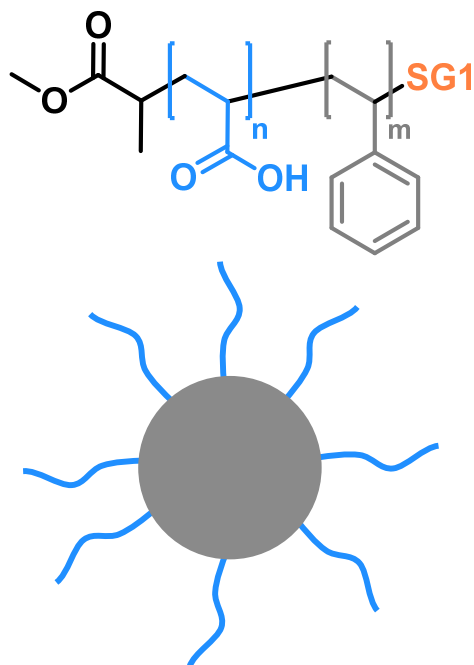


Stable Suspension



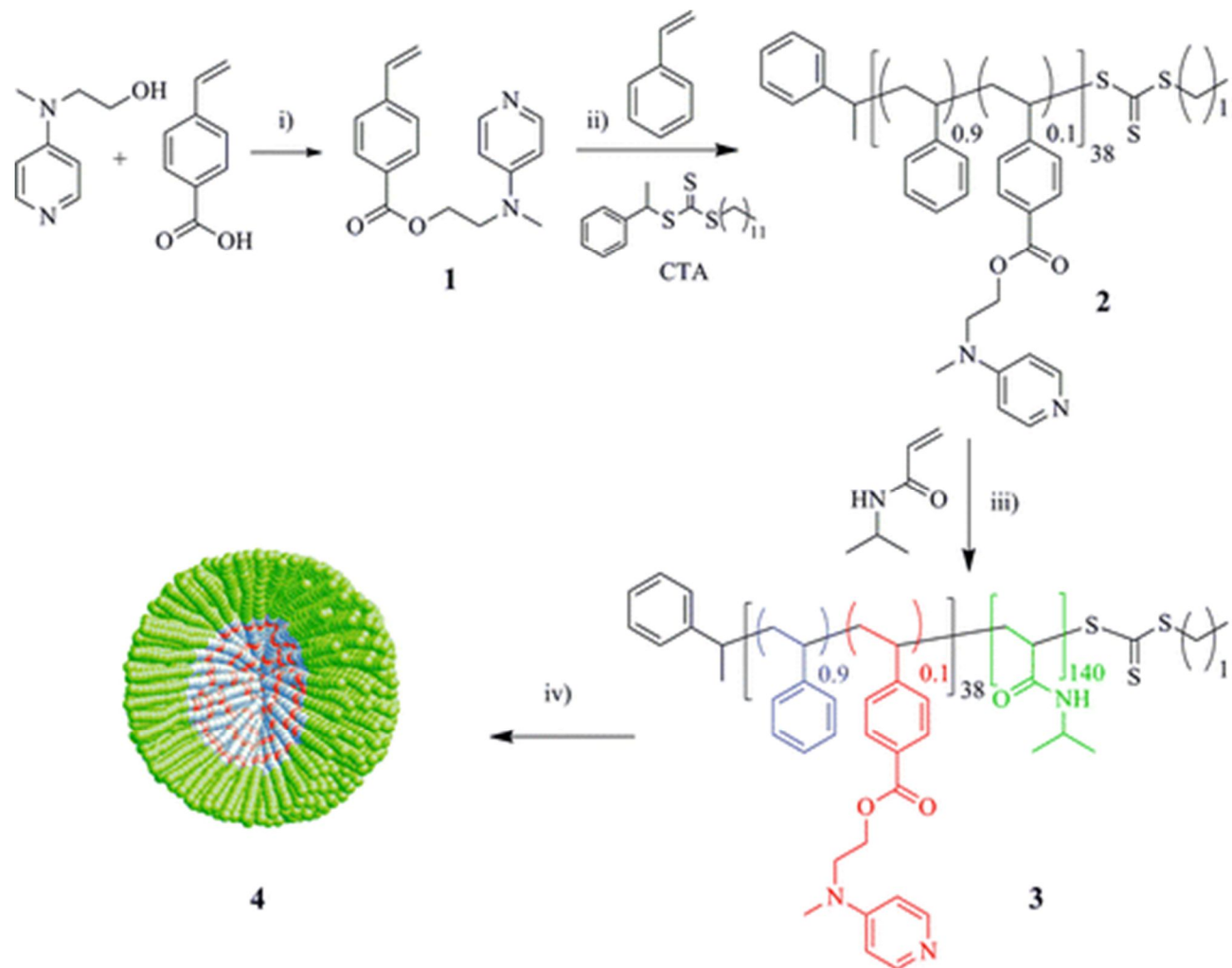
Unstable Suspension





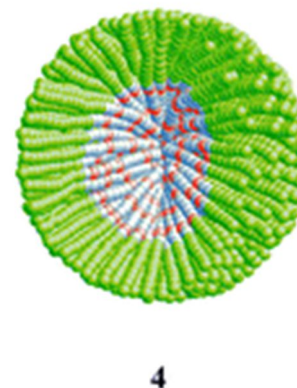
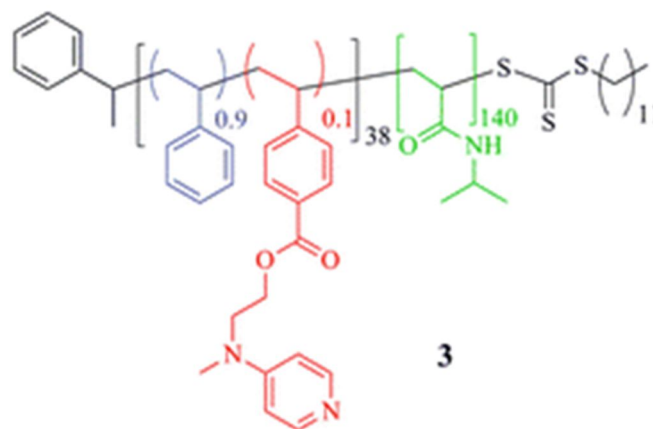
Applications of BCP Solution Self-Assemblies

Nanoreactors (core)

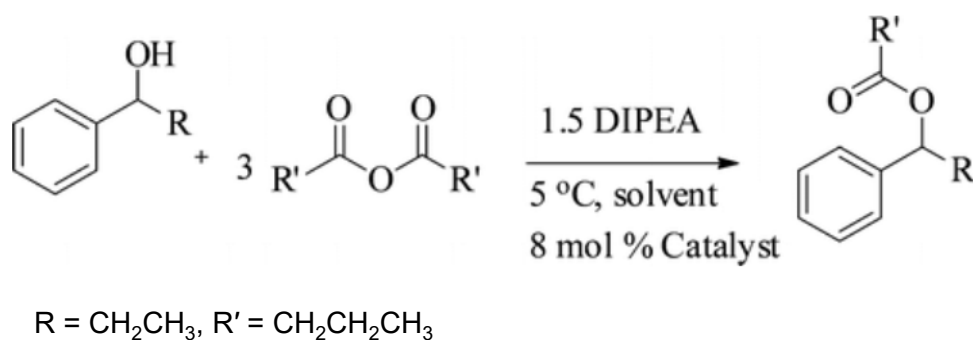


Applications of BCP Solution Self-Assemblies

Nanoreactors (core)



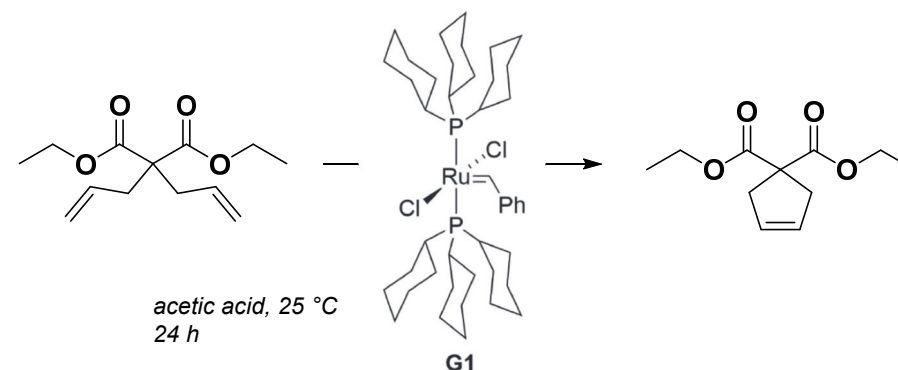
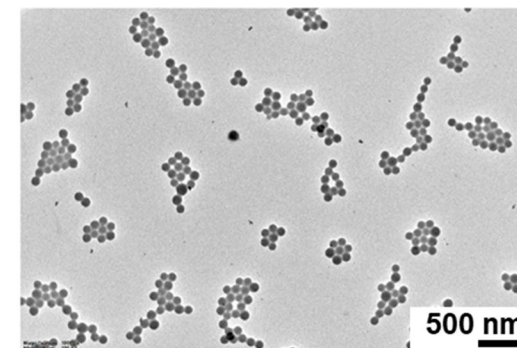
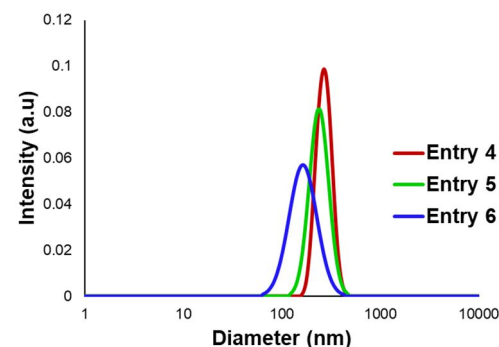
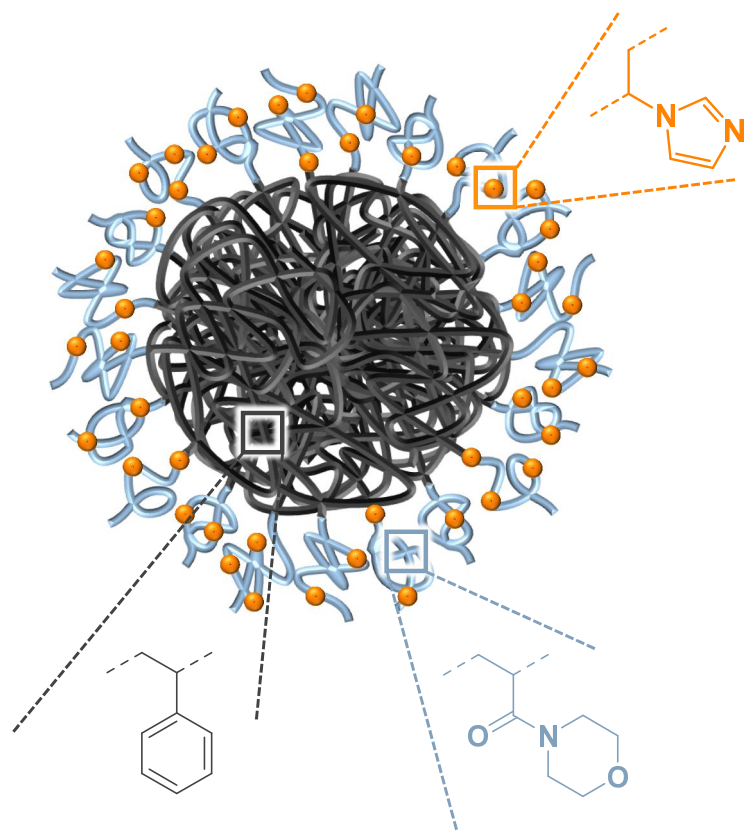
Acylation Reaction



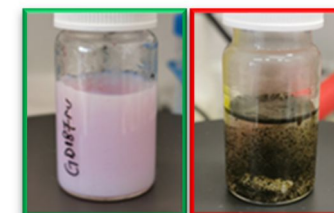
entry	catalyst	solvent	% conversion	
			15 min	24 h
1	DMAP ^b	water	2	8
2	DMAP ^b	THF	13	87
3	DMAP ^b	hexane	85	86
4	DMAP ^b	bulk	58	96
5	DMAP ^b	6^c	0	3
6	3	4^c	94	98
7	3	THF	2	60

Applications of BCP Solution Self-Assemblies

Nanoreactors (shell)



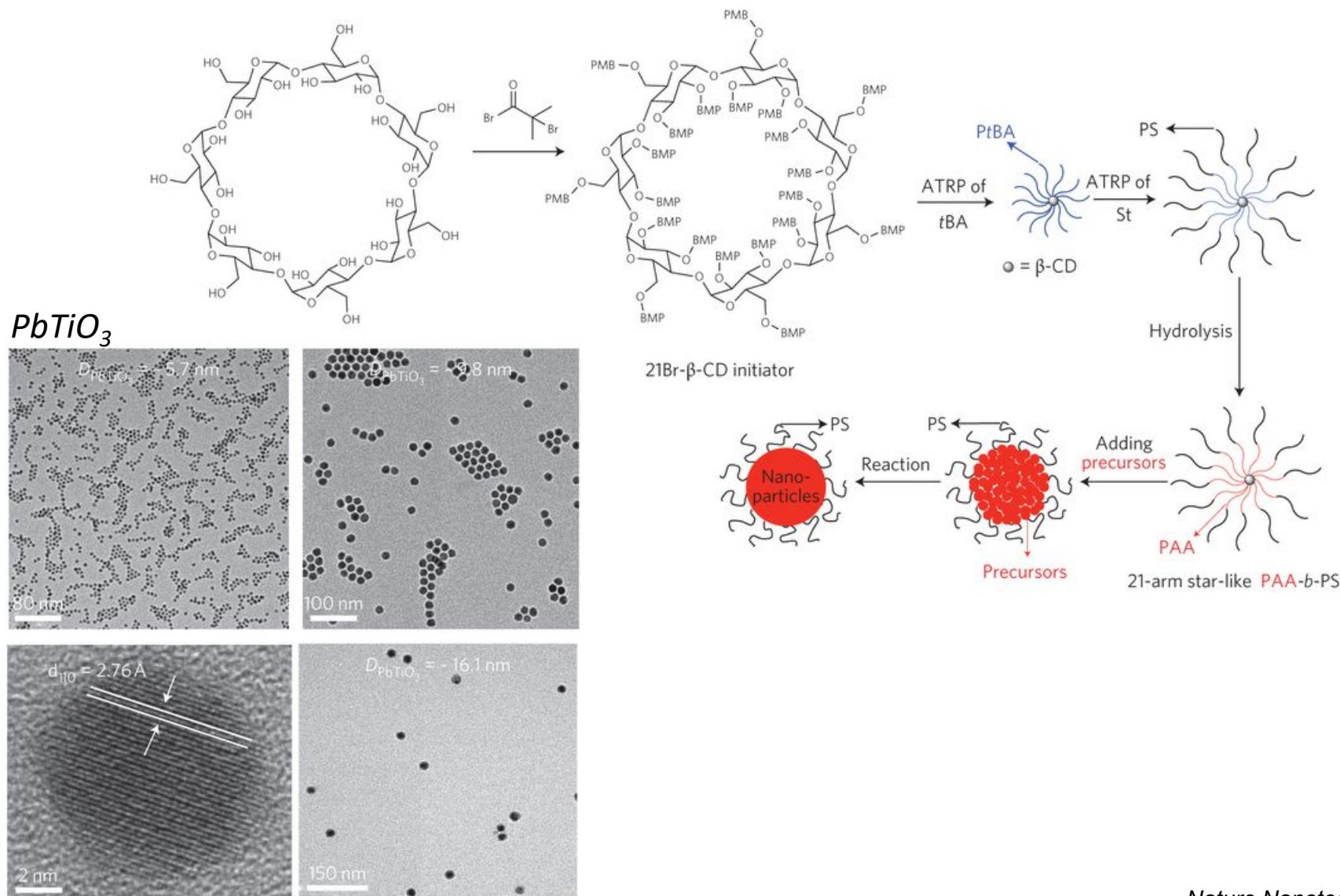
F_{VI}	Yield (%)	Ru decomposition
100	97	No
13	88	No
0	60	Yes
No NPs	60	Yes



Applications of BCP Solution Self-Assemblies

Nanotemplates

Controlling the size of inorganic particles

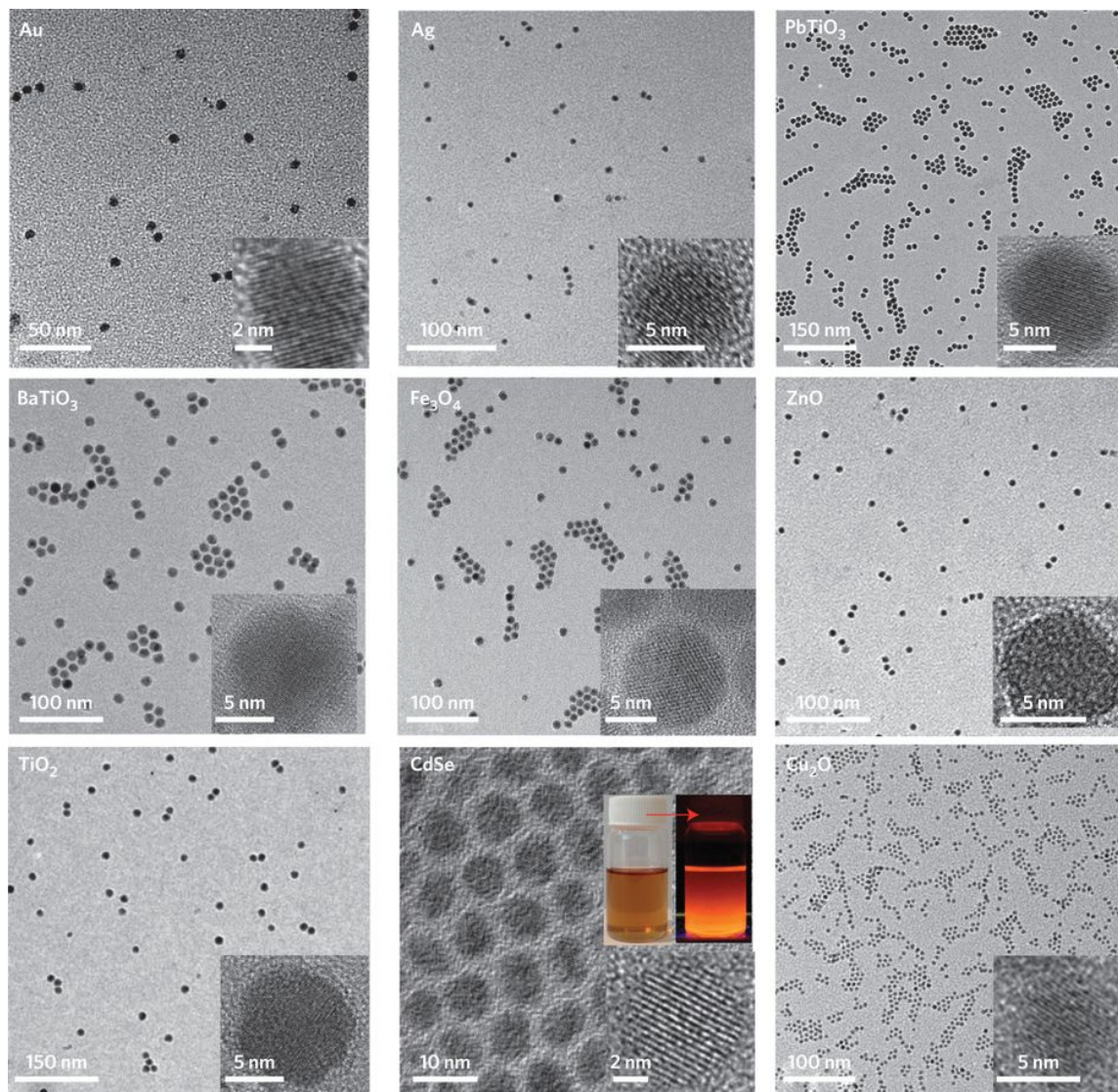


Nature Nanotechnol. 2013, 8, 426

Applications of BCP Solution Self-Assemblies

Nanotemplates

Controlling the size of inorganic particles

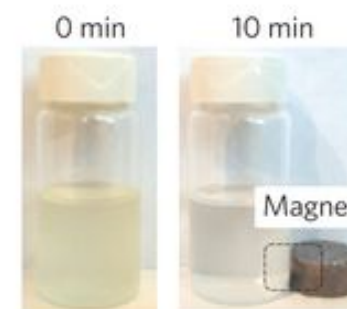
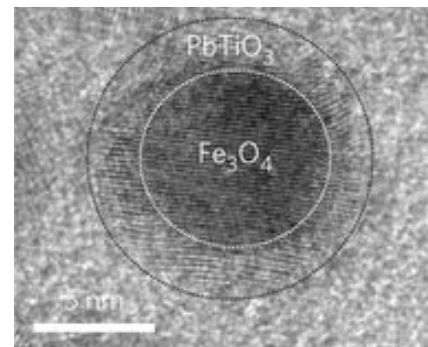
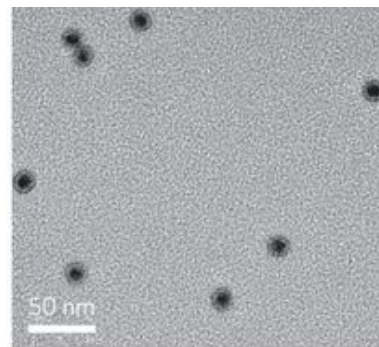
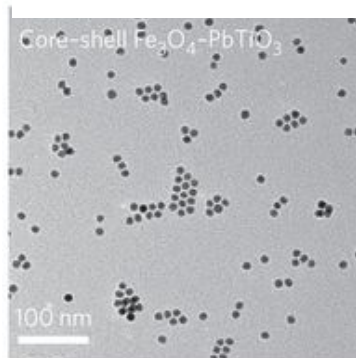
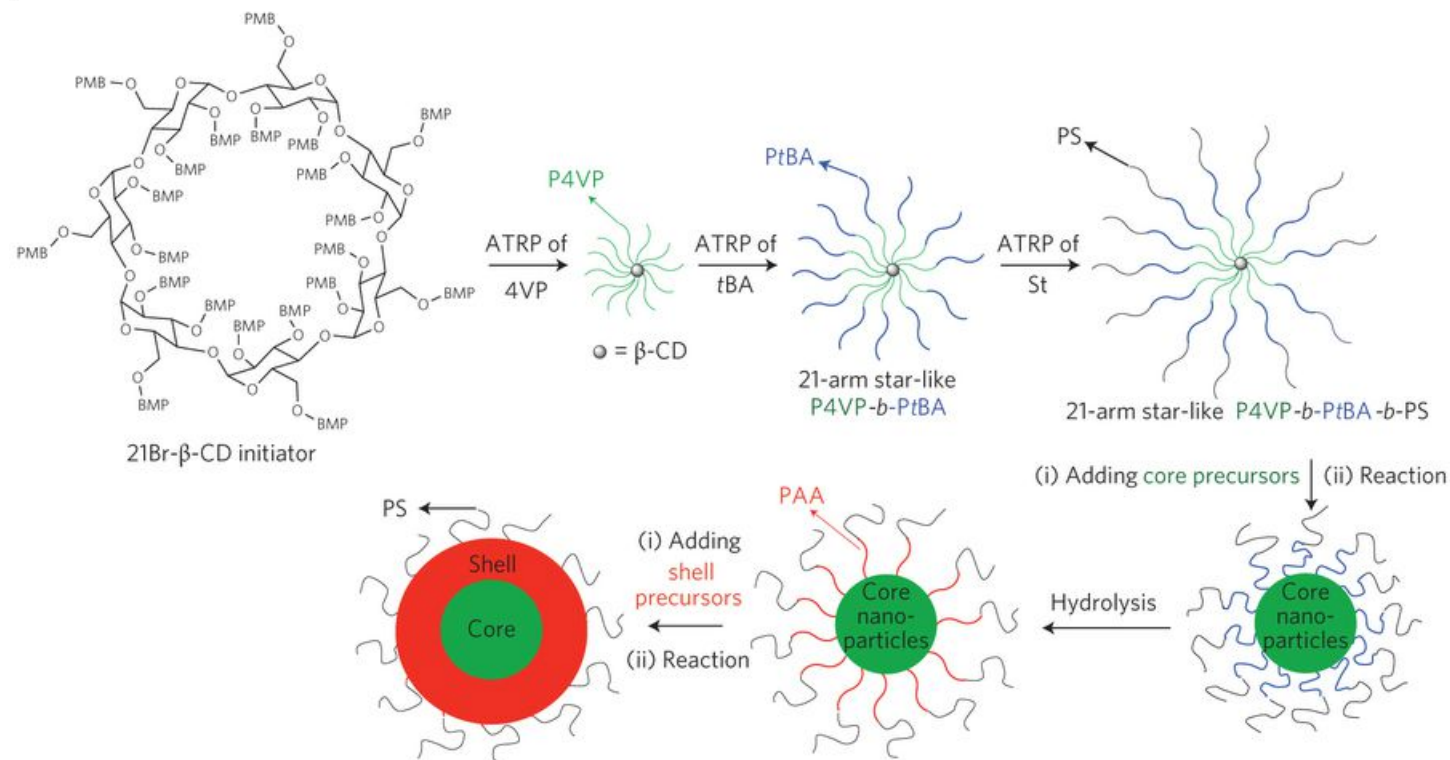


Nature Nanotechnol. **2013**, 8, 426

Applications of BCP Solution Self-Assemblies

Nanotemplates

Controlling the size of inorganic particles

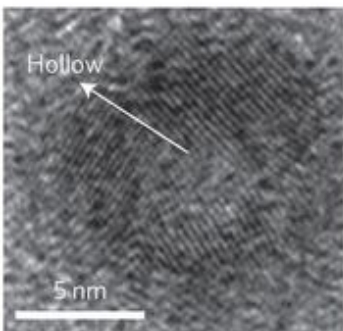
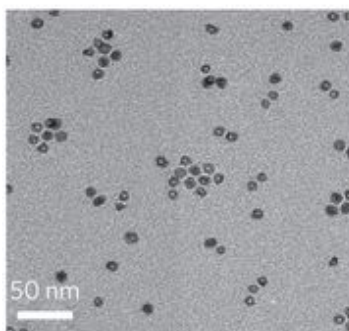
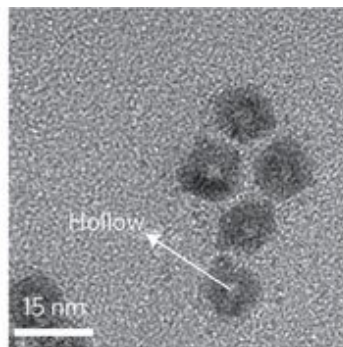
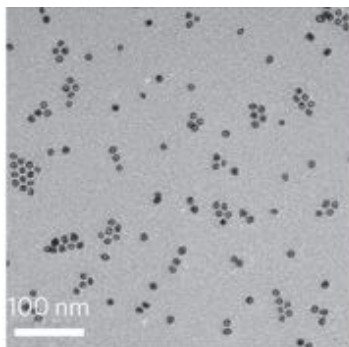
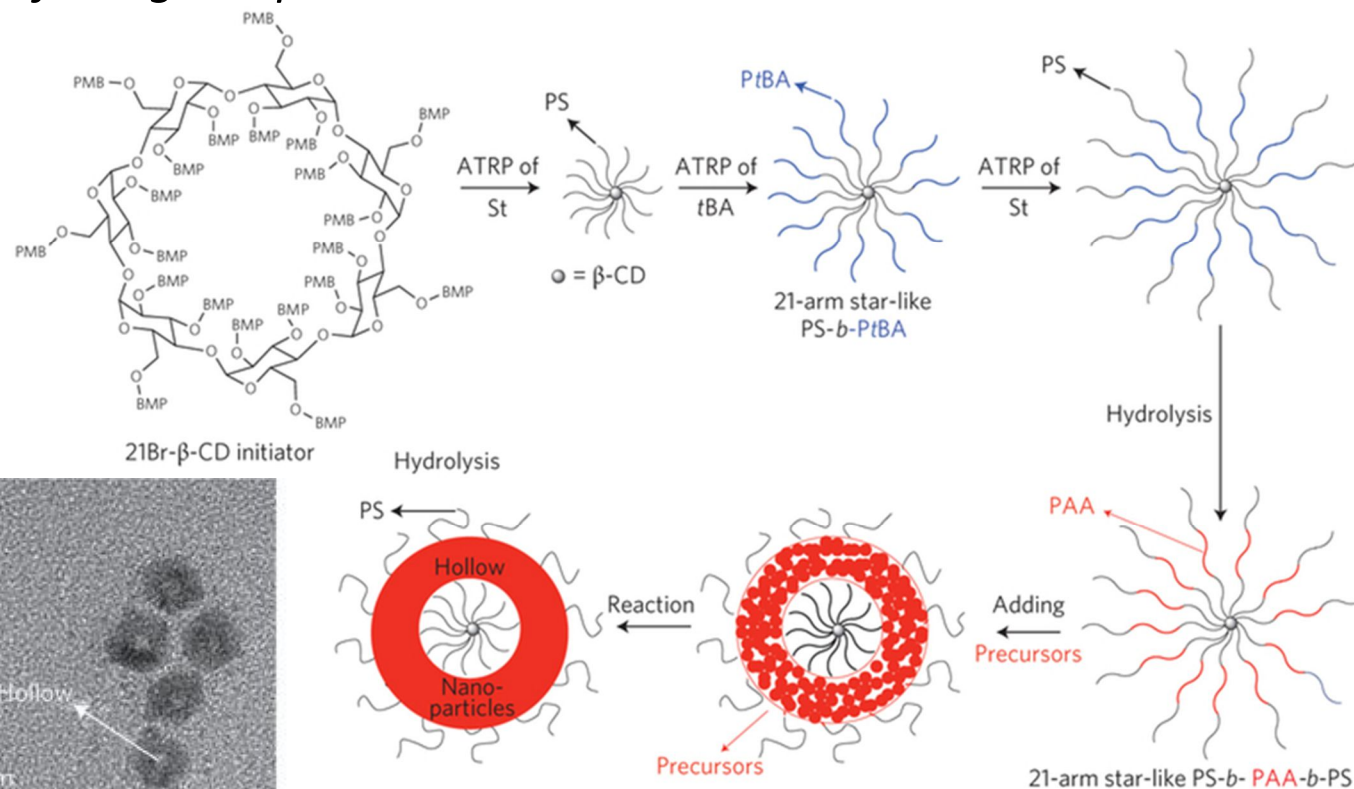


Nature Nanotechnol. **2013**, 8, 426

Applications of BCP Solution Self-Assemblies

Nanotemplates

Controlling the size of inorganic particles

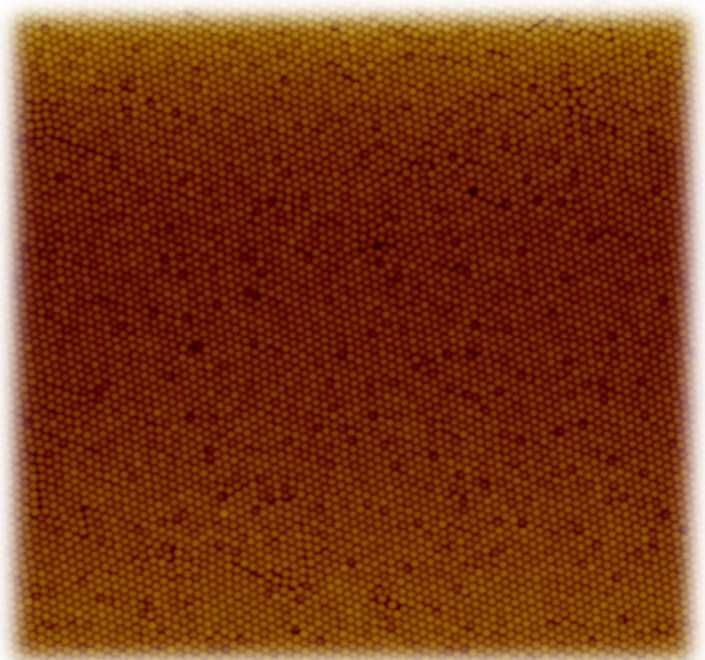


Nature Nanotechnol. **2013**, 8, 426

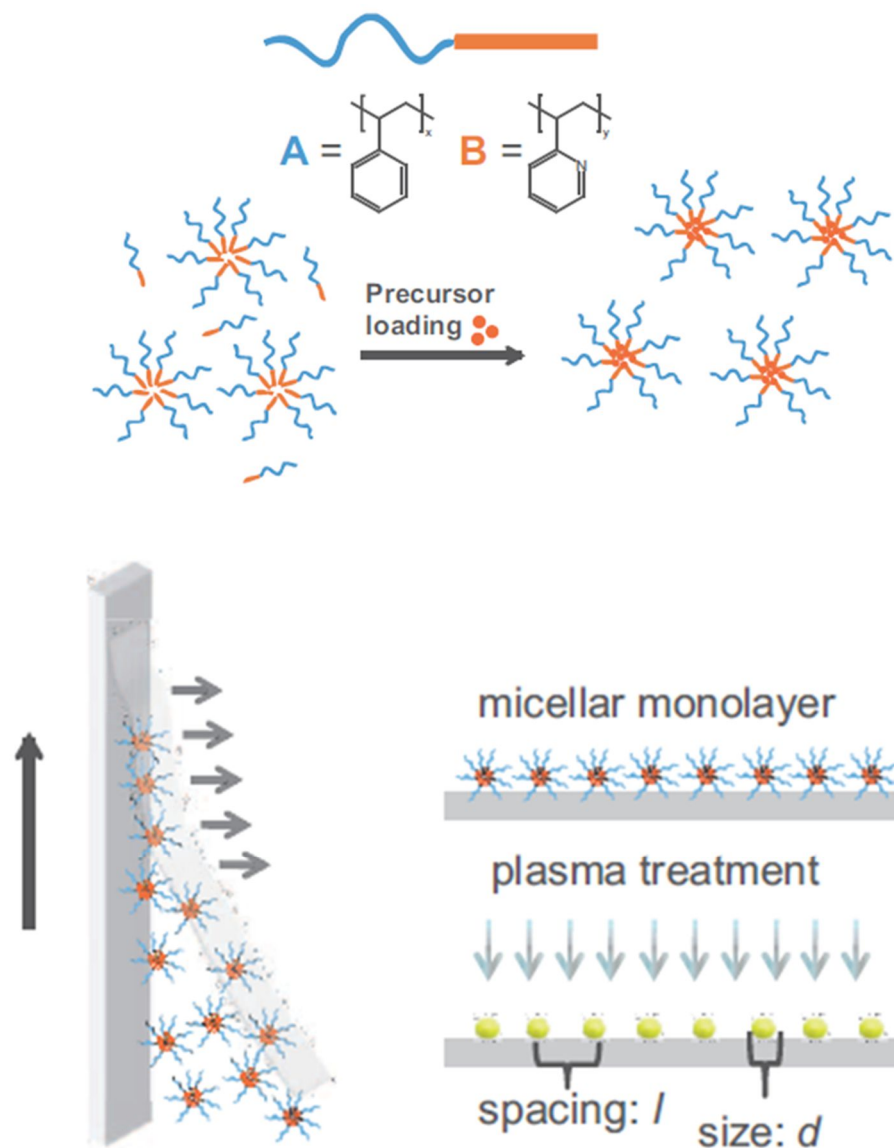
Applications of BCP Solution Self-Assemblies

Micellar Lithography

Patterning of Inorganic Nanoparticles



wikiwand.com/en/articles/Nanosphere_lithography



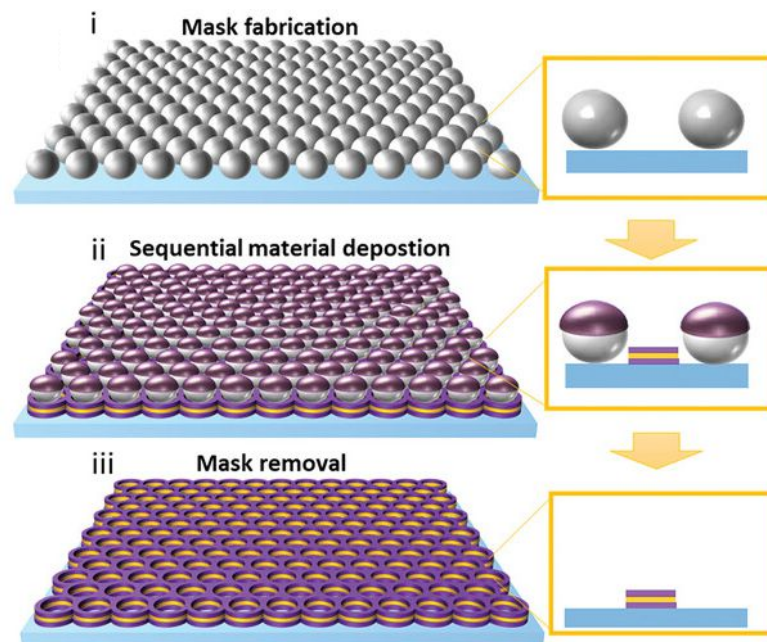
Biointerphases 2011, 6, MR1

Applications of BCP Solution Self-Assemblies

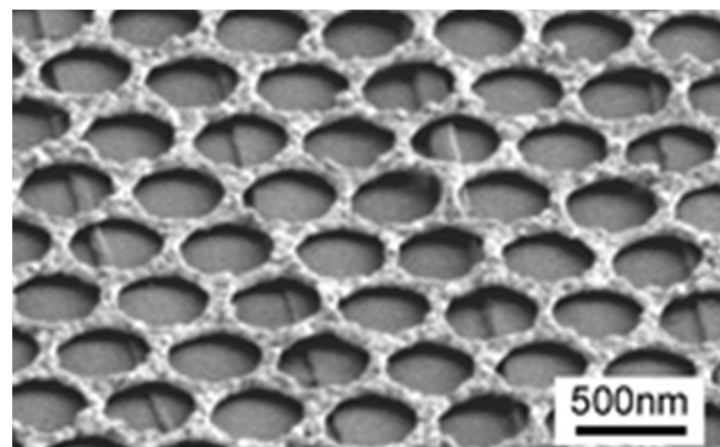
Micellar Lithography

Synthesis of transparent conductive films

Adv. Mater. 2022, 34, 2103842



wikiwand.com/en/articles/Nanosphere_lithography

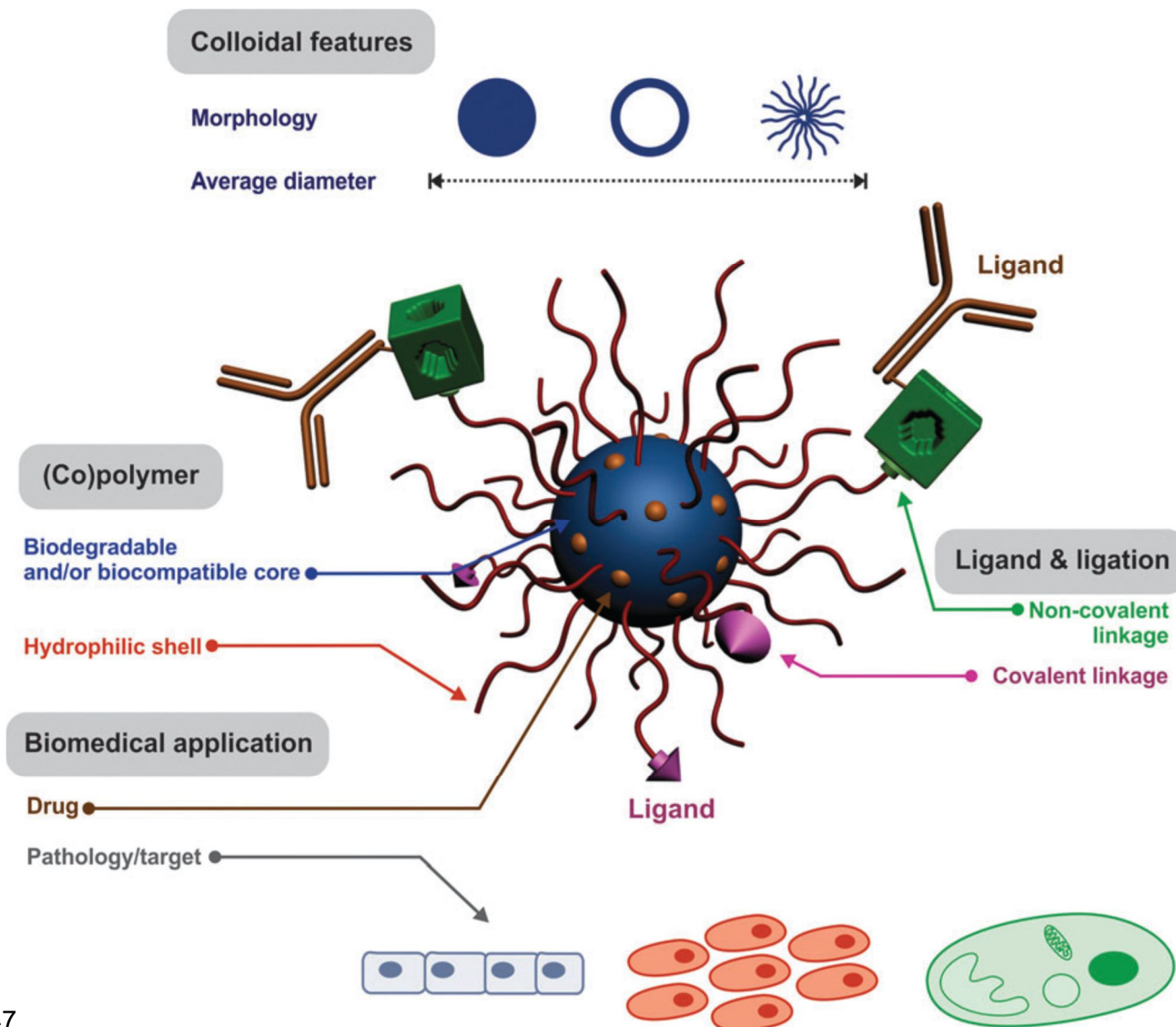


Chem. Mater. 2010, 22, 476

Applications of BCP Solution Self-Assemblies

Encapsulation/Delivery

Drug, peptide, protein, DNA, RNA

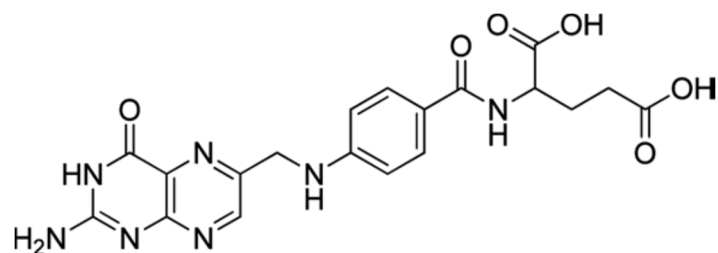


Chem. Soc. Rev. 2012, 42, 1147

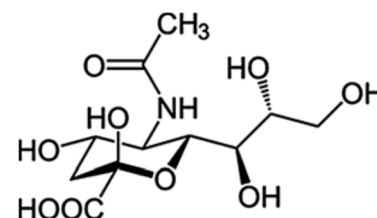
Applications of BCP Solution Self-Assemblies

Encapsulation/Delivery

Targeting Unit



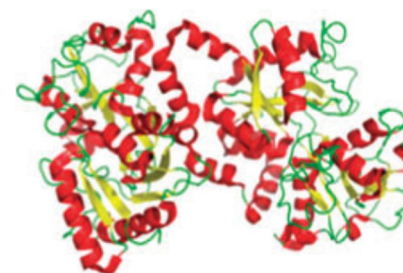
Folic acid (cancer cells)



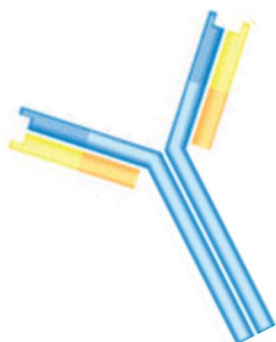
Sialic acid (CNS, central nervous system)

RGGRLSYSRRRFSTSTGR

BBB (blood-brain barrier)



Lactoferritin (brain delivery)

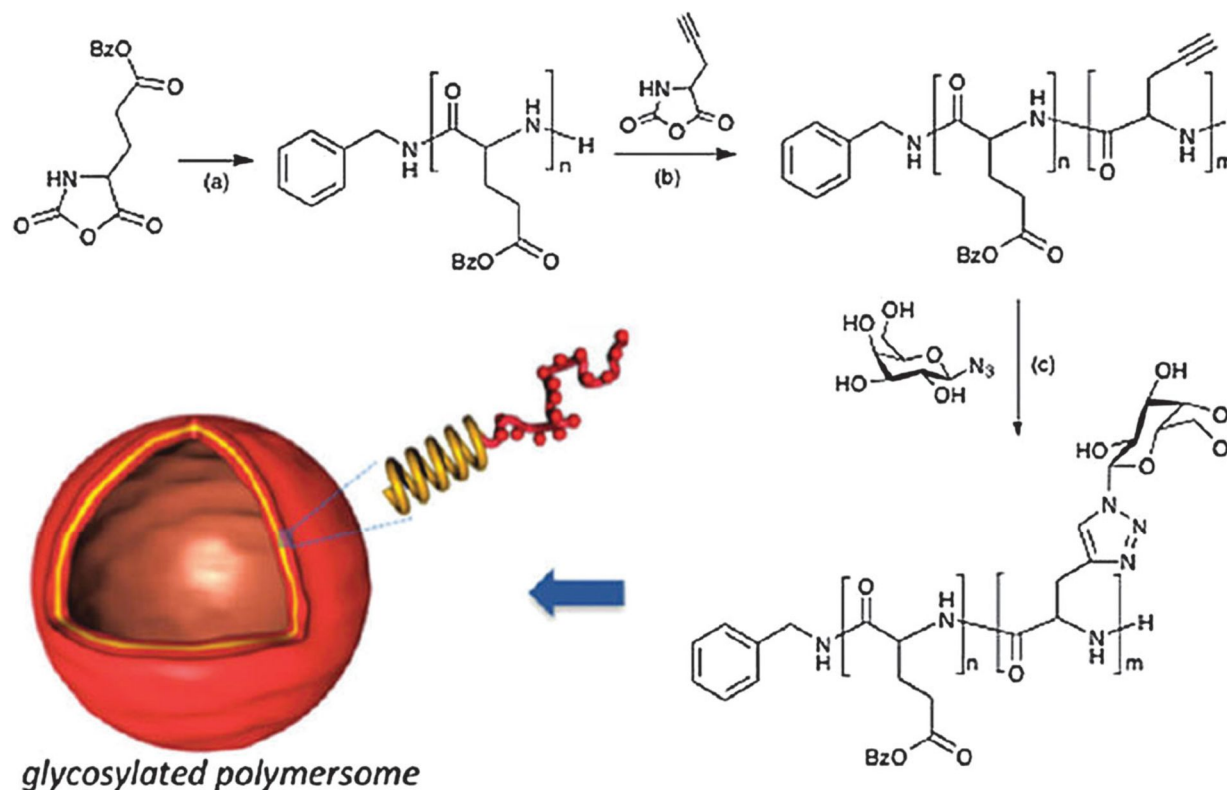


Antibodies (various antigen binding sites)

Applications of BCP Solution Self-Assemblies

Encapsulation/Delivery

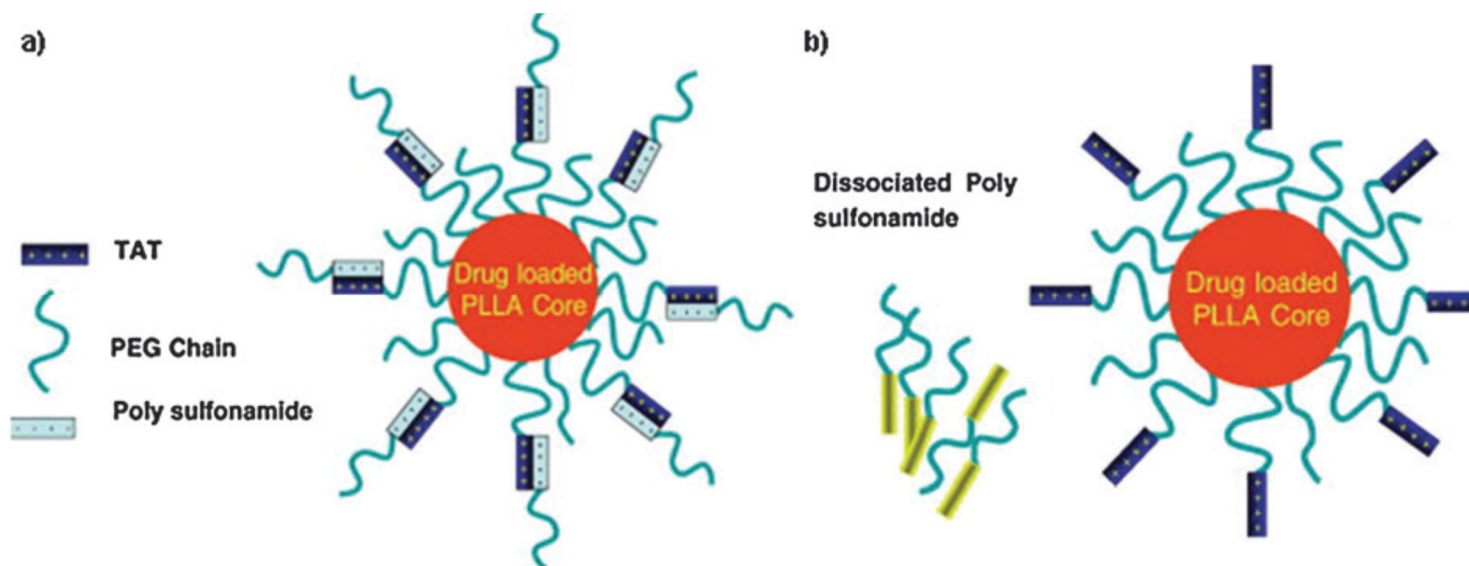
Various structures depending on hydrophilicity/hydrophobicity of the drug



Applications of BCP Solution Self-Assemblies

Encapsulation/Delivery

Responsive Structures



Encapsulation/Delivery

Radiation-Active Nanoparticles (e.g., boron-neutron capture therapy)

